

Supplemental Material

Joint High Net Load Warnings and Review Allocation under Marginal Uncertainty in Europe

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Supplemental text (Sections S1 to S15), Fig. S1, 26 CSV table exports, and ten additional supporting tables are available in the public external repository at https://github.com/williamjay1/eu-joint-netload-warnings/tree/main/supplemental_materials. The 26 exports are in `supplementary_tables/`; all files named `supporting_table_01.csv` through `supporting_table_10.csv` are in `supporting_tables/` within that repository directory. The table-export index maps files to the existing table and section labels. The number of export files is not a sequential table numbering scheme.

Supplemental methods and analyses

The main text focuses on forecast to review evaluation. This supplement supplies technical detail and complete result tables. S1 to S7 retain the earlier comparison; its original frozen test statements apply only to that earlier protocol. The subsequent original region factors and policies are reanalysis. S8 to S11 document them and the Danish transfer. S12 adds a post revision finite family check; S13 adds exploratory market/accounting context and the proposed operational interface. S14 maps tables and figures to reproducible outputs. No operator interviews or live intervention are reported.

Reading guide

Purpose	Location
Understand the monitoring target and data	S1, S2, S8, S11
Reproduce forecasts and controlled comparisons	S2 to S4, S9, S11
Evaluate allocation and measurement sensitivities	S5, S7, S10, S11
Inspect diagnostic mechanisms and uncertainty	S6, S9, S12
Assess the operating interface and new price context	S13
Locate complete tables, figures and reproduction commands	S14 and the accompanying package

Terms and units

Term	Meaning in this study
Net load	Reported load minus reported wind and solar generation, with provider specific accounting boundaries.
q90 / q95	Seasonal monitoring levels estimated from permitted previous observations. They are statistical exceedance targets, not adequacy limits.
Joint event	At least two of the three monitored areas exceed their own level in the same UTC hour.
Brier score (BS)	Mean squared difference between the issued event probability and its binary outcome. Lower is better; it measures probability loss rather than dispatch cost.
PIT	Probability integral transform: the observed value evaluated in its issued marginal CDF. It diagnoses marginal forecast calibration.
Copula	A dependence model joining marginal forecast distributions into a multivariate distribution.
ν	Student t degrees of freedom. The tested finite values change dependence shape; infinity is the Gaussian limit.
M0	Original issued margins with weather conditioned Gaussian dependence.
M1	Additional conditional marginal map with the same Gaussian dependence.
M2	Original margins with event selected, shared R Student t dependence.
M3	Additional conditional marginal map with the same selected t dependence.
P0	Compulsory top two probability hours in each complete 24 hour UTC target day window.
P1	At most two hours above or at the probability gate; zero hours are allowed.
P2	Gated risk rise priority with a cooldown and at most two selected hours per UTC day.
Review slot	One selected target hour for follow up review. It does not measure staffing time or an implemented dispatch action.
Spell / process	Consecutive active event days, with the specified allowance for inactive days; missing days split spells.
Early coverage	At least one selected true event target hour in the first six elapsed hours after realised spell onset. Forecasts were already issued; onset does not trigger a warning.
Matched expected coverage	Analytic retained hit probability after uniform, label blind sampling of the same total number of slots from each rule. This is evaluation standardisation, not a prospective annual budget controller.

S1. Observations, issue time and forecast inputs

S1.1. Outcome, source boundaries and chronology

For area j and UTC hour t , net load is $Y_{jt}=L_{jt}-W_{jt}-S_{jt}$, where L , W and S are reported load, wind and solar generation in MW. The primary event is $E_t=1\{\sum_j 1(Y_{jt}\leq h_{j,0.90}(t))\geq 2\}$. Exceedances are strict; equality is not an event. Secondary products use $q=0.95$, exceedance in all three areas, and the three named two area pairs. These pair products are projections of the same fitted three area distribution, rather than separately selected two area models.

Observations span 2019 to 2025. Numerical marginal fits begin with 2019 to 2021 history, model development uses 2022 to 2023, and final evaluation is restricted to 2024 to 2025. Exploratory 2021 diagnostics informed repairs to extrapolation, calibration and lag features and are therefore not blinded performance evidence. Earlier forward forecasts remain useful dependence training data. Historical parameters respect chronological cutoffs, but the forecasting strategy was selected with later development data; earlier forecasts are described as retrospective forward fits, not as historically issued operational forecasts. The predictive strategy, source identities, model settings and evaluation rules were frozen before test period marginal diagnostics and joint scores were inspected. Subsequent permitted parameter updates use only earlier outcomes under that strategy. A later increase in the maximum numerical integration budget resolved a warm up boundary calculation after the marginal diagnostics had been read but before any joint test score was examined. It changed neither

the probability tolerance nor the forecasting models. This chronological design is documented rather than described as a preregistered operational trial.

Electricity data come from SMARD for Germany and Luxembourg, RTE eCO2mix for France and Elia for Belgium. They are not a uniformly measured gross consumption dataset. SMARD grid load excludes pumping consumption and photovoltaic own consumption. RTE consolidated or definitive consumption covers metropolitan France excluding Corsica, includes network losses and excludes pumped storage consumption. Elia total load includes distributed load estimation, and its control area includes the Sotel area of southern Luxembourg. These provider boundaries are retained; three geographical weather polygons do not make the reported electrical quantities perfectly non overlapping national measurements.

Native intervals are preserved until aggregation. German hourly MWh are converted to hourly mean MW. French powers require both half hour records; Belgian powers require all four quarter hours. Components are aggregated separately before subtraction. Conflicting duplicates or missing components leave the affected value missing, without interpolation. Belgian historical solar records are supplemented by an archived version of the same provider product: 116,490 overlapping quarters agree exactly with current records, and 5,952 independent archived quarters agree within 0.005 MW. A missing Belgian solar quarter at local 23:45 is assigned zero only when its original neighbours both equal zero and a conservative astronomical bound is below -6° throughout the interval. This flagged reconstruction is a measurement assumption; the unchanged observed series is retained. Assigning 1 or 5 MW to a flagged quarter reduces the affected hourly net load by 0.25 or 1.25 MW relative to zero. RTE interval timestamp position is not certified by the reviewed file specification; retaining its labels is an explicit aggregation convention, with native record shifts of -30 and $+30$ minutes defined as alternatives. Consolidated labels are retrieval date versions, not reconstructed historical publication vintages.

Early Belgian photovoltaic observations were recovered from a separately documented archive of the Elia ODS032 product. This archived copy is distinguished from a newly retrieved official response; the overlap and historical export checks above support its use.

S1.2. Weather forecasts and common predictors

For target UTC day D, weather inputs use the publicly accessible NOAA GEFS initialization at D-1 00:00 UTC, and all 24 hourly warnings are issued at D-1 12:00 UTC. The 12 hour allowance is a retrospective availability policy, not an exhaustive certification of each historical object's first public arrival. Issued lead times to hour starts are 12 to 35 hours. The experiment does not evaluate warnings available before the earlier day ahead market auction. Original forecast records, member identifiers and acquisition receipts are preserved. The NOAA archive documentation identifies the operational forecast archive. Twenty perturbed members common across the archive are retained. NOAA's September 2020 system upgrade notice documents changes to the ensemble and output products; common member count and spatial sampling improve consistency but cannot remove the model upgrade itself.

Forecast endpoints at 24, 30, 36, 42 and 48 hours cover the target day. A common integer one degree node set is used: modern half degree output is subsampled at those nodes, rather than averaged into one degree cells. Regional weights use fractional polygon intersections in an equal area projection. Land polygons represent Germany plus Luxembourg, mainland France excluding Corsica and Belgium. Fixed adjacent marine boxes supply offshore weather proxies; they are neither exclusive economic zones nor installed capacity weights. Temperature and u/v wind components are interpolated to hourly midpoints at the retained nodes; hourly wind speed is then calculated at nodes before regional averaging. Downward shortwave radiation is a six hour mean flux in W m^{-2} . Solar geometry weights distribute each block's energy across hours while preserving its mean; an all zero geometry weight retains the forecast energy evenly and flags the fallback. These are forecast derived proxies and do not recover actual within block cloud variability.

The common marginal input table has 75 columns: 69 calendar, weather and response interaction terms plus six reported total load lags. Weather terms comprise ensemble means and standard deviations of temperature, land wind speed, marine wind speed and radiation in all three areas. Calendar terms include annual and daily harmonics, weekday, public holidays and slow time. Temperature and hour and renewable weather and time interactions allow demand response and changing renewable associations. Total load at exactly 14 and 21 UTC days earlier is included for each area, without interpolation. Each lag's source interval must have ended at least seven days before issue. These predictors are sufficiently old under the declared delay rule, but later provider revisions are not thereby certified historically available. The lag features enter the common marginal forecasts; they are not additional inputs to the copula correlation basis or ensemble rank templates.

S2. Thresholds, common margins and diagnostic support

S2.1. Seasonal physical thresholds and recalibration

Physical thresholds are distinct from weather conditioned forecast quantiles. At quarterly origin O , $h_{jq}(t)$ is estimated from the preceding three calendar years relative to cutoff $O-7$ days; this is a rolling interval, not necessarily three complete calendar years. Penalised quantile regression uses UTC hour indicators, two annual harmonics within four six hour blocks and block specific slow trends, with no weather predictors. Both q_{90} and q_{95} use the same design and crossing predictions are ordered. Adequate historical span and coverage are required. Calendar plus trend thresholds are selected using development single area normalised pinball losses against calendar only and fixed history candidates, without joint event frequencies or copula scores. A fixed 2019 to 2021 empirical threshold with the same UTC hour and cyclic 61 day seasonal neighbourhood is a development threshold diagnostic. The selected threshold is a monitoring convention rather than a system adequacy criterion.

For a quarter, raw conditional CDF $F_{.j}$ is fitted to history beginning in 2019 and ending before $O-7$ days- L , where L is 30 or 60 days. $F_{.j}$ is held fixed throughout the quarter. Its holdout forecasts, and subsequently its forecasts of permitted earlier days within the quarter, supply a daily calibration window. At issue $s_D=D-12$ hours, the beta CDF calibration map $G_{.jD}$ uses only intervals ending at or before s_D-7 days, within the preceding L days and no earlier than the raw fit endpoint. Thus F fitting and G fitting labels are disjoint. The first daily window can span $L-0.5$ days because the issue precedes O by 12 hours. Quarter level threshold and dependence fits use the nominal $O-7$ day cutoff, which is 6.5 days before the first issue; this clock is explicitly distinguished from the daily G cutoff. The issued CDF is $F_{.jt}(y)=G_{.jD}(F_{.j}(y))$. *Its realised forward PIT* $U_{.jt}=F_{.jt}(Y_{.jt})$ becomes training information only after the corresponding outcome is permitted.

A recorded four candidate grid crosses two raw families with matched 30/30 day or 60/60 day raw holdout and daily calibration windows. The additive location and scale model uses a Gaussian raw CDF. Its 24 weather summary terms receive cubic splines with five knots and constant extrapolation; calendar, interactions and load lags enter linearly. The scale model excludes direct slow time and its renewable weather interactions, and uses a log squared residual regression with variance correction. The penalty is 10. The alternative uses seventeen LightGBM quantile regressions between 0.005 and 0.995, with 250 trees, 15 leaves, learning rate 0.04 and minimum leaf support of 120. Reordered strictly increasing quantiles define a piecewise linear raw CDF with explicit exponential tails. Tail continuation is an extrapolation assumption. Two parameter beta calibration uses $G(u)=I_{.u}(\alpha, \beta)$, where I is the regularised incomplete beta function, fitted by prior PIT likelihood with bounded log shape parameters. CDF, survival probability and inverse quantile calculations are evaluated coherently, including small tail probabilities; numerical query clipping is not used to create incompatible marginal representations.

Marginal selection precedes dependence comparisons and uses 2022 to 2023 single area performance only. Candidates share complete UTC evaluation days and a common prior history scale per area. The score is the

equal area mean of twice the numerical integral of pinball loss on the seventeen level grid, normalised by the prior interquartile range. This is a truncated integrated quantile score rather than exact full support CRPS. Calibration checks cover PIT distributions, $q_{10}/q_{50}/q_{90}/q_{95}$, central intervals, years, seasons and forecast weather tertiles. Tertile boundaries are fitted on common earlier reference data. Pointwise intervals use 2,000 paired 14 day calendar block draws within year quarter strata, with pooled sampled sums divided by pooled sampled eligible counts; missing days remain on the lattice.

The recorded upper quantile planning envelopes are ± 0.02 at q_{90} and ± 0.015 at q_{95} . A candidate is qualified only when required pooled and weather stratum intervals lie wholly inside their envelopes and secondary checks show no supported material departure. An interval wholly outside the envelope indicates supported material failure; an overlapping or unsupported interval indicates insufficient precision. These are operational equivalence criteria, not proof of universal conditional calibration. The lowest marginal score selects among qualified candidates. No candidate qualified in development. A separate documented predictive decision adopted the additive 60/60 day specification, which ranked first in both development years, while retaining the original insufficient precision qualification and explicitly declining a calibrated margin claim. No joint event score was used in that decision. Different 30- and 60 day windows change both F and G, whereas every ensuing copula comparison retains exactly the same issued F^* values. Test period diagnostics use the frozen model and prior reference scales and weather bins, without reselection or refitting. Recalibrated denotes the fitted transformation; it does not imply that calibration equivalence has been established.

S2.2. Development selection and frozen test diagnostics

The four marginal specifications were compared on 715 common complete development days, comprising 17,160 hours. The additive 60/60 day specification had the lowest normalised truncated integrated quantile score, 0.155013, compared with 0.158440 for additive 30/30, 0.170018 for LightGBM 60/60 and 0.177804 for LightGBM 30/30. Its pointwise block interval was [0.146560, 0.157275]. It also ranked first separately in 2022 and 2023, with scores of 0.151153 and 0.158972. These scores determined a marginal predictive choice, without using joint event performance.

The candidate did not qualify under the full planned calibration screen. Among 78 unique required area/group/quantile conditions, three supported equivalence and 75 were insufficiently precise; none supported material departure, and no secondary material departure was recorded. The formal screen records each condition twice as PIT CDF and quantile coverage; their estimates and intervals coincide for these continuous monotone forecasts. Pooled upper quantile coverage was close to nominal, but the French and Belgian pointwise bias intervals were too wide to establish equivalence at either q_{90} or q_{95} . The original calibration qualified selector consequently retained no choice. A separately recorded decision adopted the best predictive candidate for the common margin experiment while preserving its diagnostic status. The selected dynamic Gaussian used the lean basis and penalty 0.001, with a 2023 q_{90} at least two selection Brier score of 0.0281116 on 353 common complete days. All fits in the completed six candidate selection grid converged. This score selects a comparator and is not independent evidence of generalisation.

The frozen marginal diagnostics retained 725 of the 731 test UTC days, comprising 17,400 hours. Four days contained known predictor unavailability and two contained a missing French observation; these exclusions were recorded separately. The fixed model's test normalised truncated integrated quantile score was 0.164904, with pointwise block interval [0.155391, 0.170122]. No test period model, reference scale or weather bin reselection was performed.

Pooled test PIT coverage at q_{90} was 0.902069, 0.885460 and 0.895345 in Germany and Luxembourg, France and Belgium, respectively; at q_{95} it was 0.950000, 0.930747 and 0.942759. Across the 78 unique required conditions, three supported equivalence, 74 remained imprecise and one supported a material departure. The departure concerns French q_{95} coverage in the high land wind speed tertile. Its identical PIT CDF and quantile

coverage records do not constitute independent findings. Coverage in that group was 0.889661, based on 3,308 hours across 275 days with group observations; the signed bias was -0.060339 , with pointwise block interval $[-0.094718, -0.017907]$, wholly below the -0.015 planning boundary. The screen entries are correlated, and their pointwise intervals do not provide simultaneous inference across all weather conditions. Most checks remain unresolved rather than positively demonstrating miscalibration. Together, these diagnostics rule out treating established conditional marginal calibration as an empirical premise. The European comparison instead evaluates dependence components using the same issued recalibrated distributions, with residual marginal uncertainty visible in its interpretation.

S3. All dependence constructions and numerical evaluation

Dependence models are updated quarterly from at most two years of earlier complete three area forward PITs, beginning in 2021 and ending at the nominal O-7 day cutoff. They use previous out of sample forecasts under the retrospective strategy, not residuals from a marginal model fitted to their own outcomes. Basis standardisation, knots and support are learned on permitted training rows. The lean correlation input table has 13 variables: annual/hourly harmonics and weekend, plus equal area mean weather and ensemble spreads for four weather quantities. The rich table adds eight between area weather contrasts. Temperature and wind and temperature and solar interactions are specified in both. Continuous terms use four knot splines with constant extrapolation.

A dynamic Gaussian copula maps three unconstrained predictors to bounded partial correlations a , b and c . The triangular factor with rows $(1,0,0)$, $(a,\sqrt{1-a^2},0)$ and $(b,c,\sqrt{(1-b^2)(1-c^2)})$ yields positive definite R through its product with its transpose. Penalised copula likelihood estimates the predictors. Six candidates cross lean/rich bases with penalties 0.001, 0.01 and 0.1. Selection uses 2023 q90 at least two Brier score on common complete days; numerical ties within 10^{-10} favour the lean basis and stronger penalty. The selected benchmark uses the lean basis and penalty 0.001. Development comparisons against it are not selection adjusted generalisation evidence.

The primary comparator C3 is that development selected dynamic Gaussian. C5 is a Student t copula inheriting exactly C3's quarterly R_t and selecting one degree of freedom from $\{3,5,8,15,30,\text{infinity}\}$ by pseudo likelihood on permitted prior PITs. Infinity is the Gaussian limit. Sharing R matches theoretical conditional pairwise Kendall dependence, $\tau_{jk}=(2/\pi)\arcsin(R_{jk})$, but not necessarily physical outcome Pearson correlation. Finite degrees of freedom change the whole copula, rather than an isolated tail. For the at least two event with unequal marginal coordinates, a t copula can increase or decrease the event probability relative to Gaussian; additional flexibility is not inherently a conservative warning correction. C5-C3 therefore measures a controlled dependence shape forecasting difference, without identifying asymptotic tail dependence or a causal weather mechanism. Shared marginal inputs remove differences in marginal model choice, but remaining marginal errors can interact differently with the two copulas and need not cancel in the comparison.

Additional distribution comparators are C1 independence, C2 static Gaussian, C4 static t sharing C2's R , and C6 dynamic t that also refits R . For each candidate degree of freedom, C6 refits R with penalised likelihood and selects the candidate using unpenalised mean copula log density on the same permitted training PITs. This empirical selection is not held out degree of freedom validation. C6 permits correlation estimation and shape changes together.

C7 constructs an ensemble rank template: an additive weather to net load mean map is fitted on one ensemble mean weather vector and realised label per historical hour, then applied to each of the 20 member aligned forecasts. Its training history follows the same beginning in 2021, at most two year restriction as the copula fits. Its separate 30 column calendar table retains the full 18 shared calendar terms and 12 holiday terms, rather

than restricting this mapping to the lean correlation basis. Total load lags are excluded from C7. Correlation modelling and C8 retain the prescribed 13- or 21 variable summary basis; C8 selects 50 permitted historical analogues by standardised distance in the transformed calendar and weather basis, including its splines and interactions, and reranks their issued PITs. Both use continuous checkerboard copulas, averaging uniform probabilities within rank cells; this gives exactly uniform component copula margins rather than a finite point quantile approximation. They receive the same recalibrated F^* as parametric competitors.

C0 is a quarterly constant event climatology: a Beta(1,1)-smoothed frequency, $(\text{events}+1)/(\text{n}+2)$, pooled over at most two years of permitted labels without hour or season conditioning. Positive slope event only recalibration of C3 (C9) and regularised direct event logistic prediction (C10) provide further decision oriented alternatives. C9 uses prior issued C3 probabilities, preserving its ordering for alert evaluation. Its monotone recalibration can alter reliability and Brier score but cannot create a genuine ranking gain. C10 uses issued marginal exceedance probabilities, their pairwise products and the common weather basis. Insufficient permitted class support is reported as unavailable.

Let $v_{jt}=F^*_{jt}(h_{jq}(t))$. With pair CDFs C_{jk} and trivariate CDF C_{123} evaluated at those coordinates,

$$p_{ge2}(t)=1-C_{12}-C_{13}-C_{23}+2C_{123},$$

$$p_{all3}(t)=1-v_{11}-v_{22}-v_{33}+C_{12}+C_{13}+C_{23}-C_{123}.$$

The corresponding pair exceedance probability is $p_{jk}=1-v_{j1}-v_{k1}+C_{jk}$. These identities evaluate each area's physical threshold under its actual conditional forecast percentile. Those percentiles need not equal q . Pair and all three outputs are consistent projections of one three area copula, preserving a common probability system rather than fitting an unrelated binary classifier for each product. Independence uses product CDFs and checkerboards use analytic cell averages. Elliptical probabilities use deterministic correlation path integration with successively increased quadrature order and declared tolerance 10^{-6} . The Student t base retains its common radial mixing: R equal to the identity does not imply independent t copula coordinates. Boundary cases use a separately checked reduced CDF procedure, with a common maximum budget of 4,194,304 points after the numerical amendment. Tighter recomputation and probability identities diagnose numerical error; convergence discrepancies are proxies, not certified bounds. An available input producing a failed probability is a failure requiring resolution, not a reason to exclude that model's difficult hours.

PIT diagnostics instead set every v_{jt} to 0.90 or 0.95 and compare probabilities with observed simultaneous forward PIT exceedances. These products distinguish upper conditional percentile behaviour from a high seasonal physical threshold that may lie in the centre of a particular weather conditioned distribution. Pooled PIT uniformity still does not establish weather conditional uniformity. Accordingly, neither a favourable physical event score nor a finite selected degree of freedom alone establishes a residual tail mechanism.

S4. Probability evaluation and complete score comparisons

S4.1. Estimand, inference, common support and reliability

The primary estimand is C5-C3 mean Brier loss, $(p_t - E_t)^2$, for physical q_{90} at least two events in 2024 to 2025. This proper event score evaluates all eligible outcomes, rather than only hours when an extreme occurred (Gneiting and Raftery, 2007). General multivariate scores can differ in their sensitivity to dependence errors (Scheuerer and Hamill, 2015); evaluation of high impact forecasts therefore needs to target the stated event while retaining an appropriate scoring rule (Allen et al., 2023). The present estimand is the named joint event probability, not an overall scenario quality score. Joint forecasts begin in 2022 to provide earlier issued C3 probabilities for C9; the evaluation window excludes that warm up period. Numerical fit provenance is checked over the entire chronology before selecting evaluation rows. Forecast eligibility depends on known feature and threshold availability, without using future outcome completeness. Observational eligibility is assessed

separately after forecasting. Every comparison uses a common set of complete 24 hour UTC days; each day receives equal weight. No model dependent missing probability mask may improve the reported comparison. Coverage exclusions and their sources are reported by year. The paired comparison changes dependence shape conditional on the recorded information set. Its interpretation remains conditional on marginal diagnostic support and on the adequacy of the available weather basis. An improved Brier score supports better forecasts of this event; it does not prove that the physical process has asymptotic tail dependence. We report the absolute score difference and relative gain, $(BS_C3 - BS_C5)/BS_C3$, alongside calibration bias and warning counts. A 3% relative Brier gain was chosen before test scores as a planning benchmark for consequential improvement; it is not an industry standard or demonstrated operator utility. An absolute difference interval is not converted into a relative gain interval by dividing its endpoints by the estimated Gaussian score, because the shared denominator is also uncertain. Relative gain uncertainty is computed by resampling both paired model scores, and the planning contrast $BS_C5 - 0.97BS_C3$ directly assesses the 3% benchmark. A score gain without fewer missed event hours at the same attention budget improves probability reporting but gives no evidence of improved ranking for that monitoring rule. Conversely, similar point estimates with wide intervals leave a consequential difference unresolved, rather than establish equivalence.

Reliability uses fixed probability edges 0, 0.005, 0.01, 0.02, 0.05, 0.10, 0.20, 0.40, 0.60, 0.80 and 1. We report bin support, mean probability, observed frequency and signed bias. Reliability diagrams depend on binning choices (Dimitriadis et al., 2021); fixed edges ensure a common comparison, and pointwise calendar block intervals retain sparse bin warnings. Paired Brier intervals use 2,000 draws of 14 day blocks within year quarter strata, retaining the calendar lattice and valid day weights. Seven- and 28 day blocks and year stratified 14 day draws assess sensitivity to temporal dependence and quarter boundaries. These intervals do not treat hours or prolonged event days as independent replications. Annual, q95, all three, pair and PIT products are secondary and their intervals are descriptive pointwise comparisons, without a claim of familywise confirmed secondary discoveries.

S4.2. Complete frozen model scores and primary contrast

The common test sample contained 2,105 physical q90 at least two event hours on 252 active days. Occurrence differed substantially between years: 1,415 of 8,712 hours in 2024 (16.24%) and 690 of 8,688 hours in 2025 (7.94%). All 16 probability products were evaluated on the same 725 complete UTC days, and their event definitions were independently reconstructed from the observation and threshold rules. These counts describe persistent hourly outcomes; they are not 2,105 independent adverse weather episodes.

The primary comparison gave essentially identical scores. Dynamic Gaussian C3 had mean Brier score 0.060190612, and its shared R t counterpart C5 had 0.060190686. The paired difference was $+7.3625 \times 10^{-8}$, with a 14 day year quarter block interval $[-1.0081 \times 10^{-5}, +7.1561 \times 10^{-6}]$. The estimated relative Brier gain was -0.0001223% , with its directly resampled paired ratio interval $[-0.0107361\%, +0.0175121\%]$. Thus, an important improvement was constrained rather than left unresolved by a wide interval. The contrast $BS_C5 - 0.97BS_C3$ was 0.001805792 [0.001554548, 0.002258865]; its positive lower endpoint excludes the preselected 3% gain benchmark at this interval level for the evaluated strategy and calendar. This benchmark assessment uses the resampled Gaussian denominator and is not obtained by dividing an absolute effect interval by a fixed estimated score. The seven- and 28 day ratio intervals were $[-0.01106995\%, +0.01709494\%]$ and $[-0.00876327\%, +0.01277334\%]$, respectively. A year stratified 14 day absolute interval also crossed zero, $[-9.3541 \times 10^{-6}, +8.5659 \times 10^{-6}]$.

The fitted dependence shape explains why this was a small intervention. C5 selected the Gaussian limit for 2024 quarter 1 and $\nu=30$ for each of the remaining seven test quarters. The correlation refitted t model C6 selected the same degrees of freedom. The static shared R t model selected $\nu=30$ in three quarters and $\nu=15$ in five. All eight selected C3 correlation fits and all eight selected C6 correlation fits converged. The

shared correlation matrices were exactly equal for the intended C4-C2 and C5-C3 comparisons throughout the 16 quarter chronology. Selection did not recover a pronounced heavier tailed conditional strategy in the primary comparison. It therefore cannot be interpreted as a successful correction of repeated tail driven missed warnings.

Table S1 places the primary contrast alongside all frozen competitors. Independence had q90 Brier score 0.061083921 and static Gaussian dependence 0.060207273, compared with 0.060190612 for C3. Adding dependence was descriptively more useful than changing the established dynamic Gaussian to the selected shared R t strategy; the improvement from static to dynamic Gaussian was itself small in this test. Weather member rank reconstruction did not improve the score, while historical analog ranks were close to C3. Event recalibration raised the q90 score to 0.061484792 but lowered the q95 score to 0.037081744; neither direction changes its within day ranking. Direct logistic prediction had a higher q90 Brier score but five more correctly selected event hours at the two hour budget. These secondary rankings distinguish probability accuracy from attention allocation and are not used to replace the primary comparison with a retrospectively selected winner.

Table S1. Mean Brier scores for all frozen competitors on 725 common complete UTC days in 2024 to 2025. Both columns concern physical seasonal thresholds and exceedance in at least two areas. These descriptive scores are accompanied by the prespecified paired C5-C3 inference in the text; the table does not establish pairwise superiority among all models.

Model	q90 Brier	q95 Brier
C0 Rolling event climatology	0.111399388	0.055556609
C1 Independence	0.061083921	0.037213114
C2 Static Gaussian	0.060207273	0.037156215
C3 Weather dependent Gaussian	0.060190612	0.037109448
C4 Static t, shared R	0.060207995	0.037147607
C5 Weather dependent t, shared R	0.060190686	0.037106669
C6 Weather dependent t, refitted R	0.060192295	0.037106117
C7 Weather member checkerboard	0.061393299	0.037695350
C8 Historical analog checkerboard	0.060277424	0.037067037
C9 Event recalibrated C3	0.061484792	0.037081744
C10 Direct logistic event prediction	0.071667514	0.045014430

S4.3. Conditional threshold position, reliability and secondary products

Physical q90 did not place each hour at the same conditional percentile. The median issued $F^*(h, q90)$ was 0.99384 for Germany and Luxembourg, 0.97377 for France and 0.97727 for Belgium. Nevertheless, 27.76%, 37.19% and 35.22% of hours, respectively, had positions below 0.90; 9.24%, 14.72% and 11.87% were below 0.50 (main article Figure 1a). A historical seasonal threshold was often far above the forecast distribution but could also lie in its middle or lower half. A predictably high level and a high conditional innovation are consequently different targets. These distributions show the positions assigned by the issued marginal model, rather than proving the corresponding conditional calibration.

Joint reliability was nearly indistinguishable for C3 and C5 (main article Figure 1b). Their mean probabilities were 0.121934 and 0.122007, against observed frequency 0.120977. Observed minus forecast mean biases were -0.000957 $[-0.013546, 0.021357]$ and -0.001030 $[-0.013622, 0.021287]$. Each fixed bin bias interval contained zero, but support and uncertainty varied widely. In the C3 probability bin $[0, 0.005)$, seven events occurred across six event days despite 8,355 forecast hours. In the highest bin $[0.8, 1]$, 471 events occupied 84 event days among 532 forecast hours, with mean prediction 0.90253 and observed frequency 0.88534.

Its bias interval was $[-0.05416, 0.02616]$. The $[0.6, 0.8)$ bin had observed frequency 0.64250 against prediction 0.69286, with bias interval $[-0.10479, 0.02416]$. All primary bins retained sufficient nonempty bootstrap support for interval reporting. Near overlapping curves and intervals crossing zero establish neither perfect reliability nor weather conditional calibration; rare bin precision is determined by event support and persistence as well as hourly counts.

The annual q90 C5-C3 point difference changed sign: approximately -2.6303×10^{-6} in 2024 and $+2.7850 \times 10^{-6}$ in 2025. Raising the physical threshold to q95 reduced occurrence to 1,001 hours on 155 active days and yielded C3 score 0.037109448, C5 score 0.037106669 and difference -2.7787×10^{-6} $[-9.3875 \times 10^{-6}, +1.2488 \times 10^{-6}]$. All three and pair projections are reported in Table S2, alongside the PIT diagnostics. Some physical all three and pair intervals excluded zero in favour of C5, but differences were only several millionths of a Brier unit. They are dependent secondary pointwise comparisons without multiplicity control, and numerical convergence diagnostics do not provide a strict paired score error bound at that scale. They neither overturn the primary result nor identify a physical tail mechanism. The four fixed PIT contrasts all crossed zero. Their event frequencies also differed from physical threshold frequencies, as expected for different outcomes.

Table S2. C5-C3 contrasts for all physical and PIT event products on the same 725 complete days. Differences and intervals are in units of 10^{-6} Brier score; negative values favour C5. Intervals use 2,000 paired 14 day year quarter block draws. The first row is primary; all remaining rows are descriptive secondary pointwise comparisons. Pair abbreviations refer to Germany and Luxembourg (DE.LU), France (FR) and Belgium (BE).

Threshold and event	Event hours	Active days	Difference $\times 10^{-6}$	Pointwise 95% interval $\times 10^{-6}$
Physical q90, at least two	2,105	252	+0.0736	$[-10.0811, +7.1561]$
Physical q90, all three	566	103	-5.8218	$[-8.9785, -2.7047]$
Physical q90, DE.LU-FR	924	143	-3.4979	$[-6.6622, -0.1229]$
Physical q90, DE.LU-BE	1,150	172	-0.3782	$[-5.9939, +4.2884]$
Physical q90, FR and BE	1,163	165	-4.8293	$[-8.9100, -0.7320]$
Physical q95, at least two	1,001	155	-2.7787	$[-9.3875, +1.2488]$
Physical q95, all three	203	52	-3.2021	$[-7.0184, -0.5598]$
Physical q95, DE.LU-FR	408	83	-2.5332	$[-6.7630, -0.2277]$
Physical q95, DE.LU-BE	512	92	-2.6199	$[-5.1905, -0.4946]$
Physical q95, FR and BE	487	94	-2.6408	$[-7.1402, +0.2930]$
PIT 0.90, at least two	1,109	264	-12.2450	$[-38.2997, +20.4138]$
PIT 0.90, all three	268	91	-1.2318	$[-7.9970, +5.2478]$
PIT 0.95, at least two	520	149	-10.4766	$[-24.2262, +6.2153]$
PIT 0.95, all three	106	41	-0.5075	$[-2.9471, +1.5539]$

The mathematical projections derive from one joint distribution, but numerical calculations have finite residuals. Across the physical probability products, the largest absolute residual in $\text{sum}(\text{pair probabilities}) - p_{\text{at-least-two}} - 2p_{\text{all-three}}$ was 3.40×10^{-6} , and the largest containment order violation was 3.52×10^{-6} . For primary q90 C5 these were 2.01×10^{-6} and 9.61×10^{-7} . These checks distinguish mathematical coherence from exact floating point consistency. The 10^{-6} integration criterion is a convergence proxy, not a universal bound on projected probabilities or model score differences. No substantive finding is attached to the sign of the near zero primary estimate.

S5. Daily review capacity, persistent events and alert sets

S5.1. Allocation and event grouping

A daily attention budget ranks the next 24 hourly probabilities at issue and selects the highest two; one- and four hour budgets assess capacity sensitivity. UTC order resolves ties. A whole period ex post top percentage is not used. Hourly miss rate, false alert fraction among alerts and false positive rate among non events are separately reported. C9's ranking explicitly reuses C3, preserving equality even if numerical saturation creates ties in its transformed probabilities. Improved probability calibration is thus distinguished from better allocation of a fixed warning budget. The daily ranking consumes issued hourly probabilities only. Neither a sum nor a maximum of those probabilities is interpreted as the probability of any event during the day, because the forecast does not model joint dependence across the 24 horizons.

An event spell combines consecutive active days and bridges at most one fully observed inactive day; zero- and two day bridges assess sensitivity. Missing days break spells and flag uncertain boundaries. This is an event based persistence proxy, not a certified independent meteorological episode. We record any hit, first day hit, and an alerted event within the first six elapsed hours after onset. The latter does not mean the first six event containing hours. Issue to target and issue to onset lead times are distinct. Spell rates are reported with support and boundary status, without an independent binomial precision claim.

The allocation uses exactly K slots on each complete day, including days without events. The selected hours are candidates for review, not an observed dispatch action or a probability threshold alarm. For realised daily event counts n_d , the perfect information limit is $\sum_d \min(n_d, K)$. It is computed after evaluation to explain the capacity constraint and cannot be implemented from forecast information.

S5.2. Observed detection, descriptive capacity and alert set identity

At the two hour daily budget, C3 and C5 each selected 1,450 alert hours, with 341 true positives, 1,764 missed event hours and 1,109 false alerts. Their miss rate was 83.8005%, false alert fraction 76.4828% and false positive rate 7.2507%. Paired differences for all three rates were zero with $[0, 0]$ block intervals. The models exchanged 25 selected hours each across 19 days, giving a symmetric difference of 50 alert hours without an aggregate detection gain. Equality of these evaluated counts and rates therefore does not imply equality of selected hours or forecast distributions. At one alert hour per day both models detected 179 event hours; at four, C3 detected 667 and C5 666. The q95 two hour counts were also equal: 188 detected events, 813 misses and 1,262 false alerts.

The high hourly miss percentage partly reflects the chosen capacity on persistent event days. A post evaluation perfect information calculation gives an upper limit of $\sum_d \min(\text{event hours on day } d, 2) = 476$ q90 event hours that could be selected under this budget, even with known future labels. Thus, at least 77.3872% of event hours would remain unselected under perfect information, and the observed 341 hits represent 71.6387% of this capacity. The analogous q95 upper limit was 290 hours, with C3 and C5 attaining 64.8276%. This ex post descriptive bound is not deployable skill or a prespecified utility threshold. It explains why hourly recall alone cannot measure whether persistent adverse periods received any warning.

With the main one inactive day bridge, 252 q90 active days formed 71 event spells, of which 68 had certain boundaries and three had uncertain boundaries. C3 and C5 each detected at least one event hour in 65 of 71 spells (91.55%), on the first active day in 50 (70.42%), and within the first six elapsed hours after onset in 35 (49.30%). Restricted to the 68 spells with certain boundaries, these counts were 62 (91.18%), 48 (70.59%) and 34 (50.00%). Eventual spell coverage therefore concealed weaker coverage near onset: more than nine in ten spells included an alerted event hour, but only about half included one in the first six elapsed target hours. All warnings for a target day were issued together at 12 UTC on the preceding day. The onset statistic

measures overlap between selected target hours and realised onset, rather than a notification or publication delay.

Changing the bridge from zero to two observed inactive days produced 93 and 51 descriptive q90 spells, with C3/C5 any hit rates 87.10% and 96.08% and first six hour rates 54.84% and 50.98%. All 93 zero bridge spells had certain boundaries; 47 of the 51 two day bridge spells had certain boundaries. First day coverage was 67/93 (72.04%) and 36/51 (70.59%), respectively. For q95, the main bridge produced 61 spells, including 60 with certain boundaries; C3/C5 any hit, first day and first six hour rates were 86.89%, 65.57% and 47.54% over all 61. These grouping dependent summaries are not independent weather system replications. They reveal limited target hour coverage near onset that the selected dependence shape strategy did not repair.

S6. Controlled simulation and diagnostic results

S6.1. Generating mechanisms and independent fitting blocks

Five known generating mechanisms each have 100 independent repetitions. Each repetition uses four disjoint ordered blocks: 2,000 correlation fit rows, 1,000 degree of freedom selection rows, 1,000 event recalibration rows and 1,000 new test rows. Weather X is equiprobable $-1/+1$; true conditional margins are normal with means $X(0.4, 0.3, 0.2)$ and unit scale. Copula draws U map to outcomes through $Y_j = \mu_j(X) + \Phi_{\text{inverse}}(U_j)$, so changing the copula does not change the normal outcome margins. Partial correlation predictors are $\eta(X) = (0.55, 0.45, 0.30) + X(0.25, 0.20, 0.15)$, mapped to R by the same triangular construction as described in S3. The omitted state scenario adds $H(0.65, 0.65, 0.65)$ to η , where H is independently equiprobable $-1/+1$. Mechanisms are dynamic Gaussian, matched R t5, matched R t8, a mixture of Gaussian structures caused by an omitted independent weather state H , and the Gaussian case evaluated with erroneous margins of mean bias $+0.25$ and scale 0.8 . The event is at least two outcomes above $\Phi_{\text{inverse}}(0.90)$.

Dynamic Gaussian fitting uses penalty 0.001 . The t procedure inherits its exact R and selects degrees of freedom on the separate second block; this disjoint simulation selection differs from the empirical permitted history pseudo likelihood fit. Event recalibration uses the third block only, with penalty 0.01 and a nonnegative slope. Known DGP conditional probabilities support probability RMSE checks, and restored H and correct margin Gaussian controls diagnose missing information and marginal misspecification. They are not same information algorithm competitors. Paired mean Brier intervals use repetition level standard errors across 100 independent repetitions. Formal integration tolerance is 10^{-4} , with repetitions 1 to 10 per mechanism predetermined for 10^{-5} recomputation. All attempted units are retained. A post design prior block PIT audit adds q10/q50/q90/q95 and an IID empirical CDF warning without changing models or test results. Its IID concentration bound is applicable to this synthetic experiment, not unadjusted European weather series.

S6.2. Results for all 500 repetitions

All 500 repetitions completed without failed attempts. Table S3 shows the shared R selected t comparison. Under t5, conditional probability RMSE decreased from 0.006012 to 0.001132 and the mean Brier difference favoured t by 2.216×10^{-5} . Under t8, RMSE decreased from 0.003949 to 0.001127, but the smaller Brier difference remained imprecise. The selected degree of freedom equalled 5 in 93 t5 repetitions and 8 in 87 t8 repetitions. Matching the correlation structure allowed the flexible copula to recover a known finite threshold probability mismatch, although the effect on the realised Brier score depended on its magnitude and sampling variability.

Table S3. New test results for synthetic mechanisms, with 100 paired repetitions each. Differences are selected shared R t minus dynamic Gaussian; negative values favour t. Intervals describe Monte Carlo uncertainty in the repetition mean difference and do not describe European weather episodes.

Generating mechanism	Gaussian Brier	shared R t Brier	Mean difference	95% Monte Carlo interval	Finite degrees of freedom
Dynamic Gaussian	0.06697900	0.06698039	+0.000001392	[+0.000000283, +0.000002500]	20/100
matched R t5	0.07190664	0.07188449	−0.000022155	[−0.000035903, −0.000008408]	100/100
matched R t8	0.07008500	0.07007885	−0.000006149	[−0.000015168, +0.000002869]	100/100
Omitted Gaussian weather state	0.06119732	0.06131180	+0.000114475	[+0.000086763, +0.000142187]	100/100
Erroneous margins, Gaussian copula	0.06704282	0.06704420	+0.000001376	[−0.000001232, +0.000003984]	92/100

The omitted weather mixture selected degree of freedom 3 in every repetition, yet t increased probability RMSE from 0.006503 to 0.011365 and signed event probability bias from +0.002379 to +0.012188. Restoring H in a Gaussian model reduced Brier score relative to the X only Gaussian by 0.0011493, with interval $[-0.0012662, -0.0010324]$. This improvement uses additional information. Its size and direction show why a fitted heavier tailed dependence parameter cannot substitute for diagnosing an omitted conditioning state.

Erroneous margins produced finite degrees of freedom in 92 repetitions even though the generating copula was Gaussian. Correcting the known margins on the same outcomes reduced Gaussian Brier score by 0.00006384 $[-0.00008950, -0.00003817]$. Gaussian event recalibration increased mean Brier score in all five scenarios. Recalibrating t improved its own forecast in the omitted state scenario but still left it worse than uncalibrated Gaussian. Event correction consequently neither guarantees improved new sample prediction nor establishes a correct joint distribution.

The additional prior margin audit explains a misleading upper quantile check. For the wrong margin, $0.25 + 0.8\Phi_{\text{inverse}}(0.90) = 1.27524$ is close to the true 1.28155; theoretical PIT coverage at q_{90} is 0.898888. Empirical coverage averaged across areas was 0.21877 at q_{10} , 0.59883 at q_{50} , 0.89822 at q_{90} and 0.94105 at q_{95} . The original q_{90} only warning flagged none of these 100 repetitions, whereas the full prior PIT CDF warning flagged all. Full CDF warnings also occurred in 4 to 7% of scenarios with correct margins, consistent with finite sample diagnostic error. Broader diagnostics therefore expose this construction without becoming perfect classifiers.

The correctly specified Gaussian scenario selected infinity in 80 repetitions and 30 in twenty. Its tiny positive t minus Gaussian Brier difference has no substantive warning interpretation. Predetermined tighter integration changed probabilities by at most 9.0201×10^{-5} and the mean test Brier score of a model within an audited repetition by at most 1.2472×10^{-6} . The latter averages 1,000 test hours; it is neither an individual hour loss bound nor a certified bound on a paired or pooled contrast. The simulation supports distinguishing a finite threshold copula mismatch from missing weather information or wrong margins. These findings concern the specified synthetic generating mechanisms and do not establish residual tail dependence or under warning in the European observations.

S7. Measurement conventions and fixed forecast sensitivity

S7.1. Matched domain observation checks

Measurement sensitivity holds the issued probabilities and main physical thresholds fixed. Belgian flagged solar quarters are assigned 1 or 5 MW instead of zero, and strict observation evaluation removes reconstructed hours or incomplete days under the same common support rule. French alternatives shift the native records by -30 or $+30$ minutes before complete hour aggregation. Event labels, scores and alert outcomes are then recalculated on matched eligible observations; sample exclusions are shown separately from changes in performance. These are label only checks of the chosen observation convention, not evidence that re-estimating the entire forecasting system under an alternative measurement model gives the same result. Annual, event definition, PIT, budget, block, spell and measurement comparisons are secondary pointwise analyses under the frozen evaluation. The fixed 2019 to 2021 threshold remains a development diagnostic.

S7.2. Scores, label changes and sample restriction

All ten label only evaluations completed under the five specified measurement alternatives. Table S4 reports the primary contrast for both physical thresholds. Assigning 1 or 5 MW to flagged Belgian night time solar quarters changed no known event labels, retained all 725 complete days and left every model's Brier score unchanged. Strict observation evaluation retained only 437 common complete days, excluding 288

original days. It changed no known labels. On that common domain, original and strict observation C5-C3 score differences were identical, so the entire change from the full sample contrast was attributable to day restriction rather than altered event classification. A nominally stringent observation rule therefore changed which periods were evaluated without demonstrating a different warning process.

The French -30 minute convention retained 722 complete days and changed 250 q90 and 163 q95 hourly labels; +30 minutes retained all 725 days and changed 222 and 148 labels. All changed labels fell within their common complete day domains. The q90 common day label contribution to the contrast was -2.3207×10^{-6} for -30 minutes and $+0.9757 \times 10^{-6}$ for +30 minutes. The former also had a small day restriction contribution of $+0.0346 \times 10^{-6}$. All ten official 14 day contrast intervals crossed zero. These results show that the near zero comparison persisted for fixed issued forecasts under the specified alternatives. They certify neither a physical timestamp interpretation nor robustness of a completely refitted pipeline under different measurement assumptions.

Table S4. Fixed forecast C5-C3 Brier differences under alternative observation conventions. Columns show differences and 14 day year quarter pointwise intervals in units of 10^{-6} , and complete day support. Main thresholds and issued probabilities are unchanged; all analyses are descriptive secondary checks.

Convention	Complete days	q90 difference and 95% interval $\times 10^{-6}$	q95 difference and 95% interval $\times 10^{-6}$
Strict observed records	437	+3.5351 [-9.1528, +14.5286]	-0.5414 [-7.5664, +4.1683]
Flagged night solar assigned 1 MW	725	+0.0736 [-10.0811, +7.1561]	-2.7787 [-9.3875, +1.2488]
Flagged night solar assigned 5 MW	725	+0.0736 [-10.0811, +7.1561]	-2.7787 [-9.3875, +1.2488]
French timestamps -30 minutes	722	-2.2125 [-11.6788, +5.0854]	-1.0954 [-7.7150, +3.9629]
French timestamps +30 minutes	725	+1.0493 [-9.2548, +7.2064]	-2.0340 [-8.9275, +2.3066]

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S8. Development audit and information clocks

Twelve calendar-fixed development weeks cover winter, summer and both clock changes in 2022 and 2023. They contain 2,016 scheduled UTC hours, 2,014 jointly complete hours and 82 complete UTC days. The two French gaps are absent in the acquired raw RTE rows at 00 UTC on 30 October 2022 and 29 October

2023. Adjacent half-hours convert correctly; UTC modelling days remain 24 hours. This traces the gaps to the retrieved source snapshot without determining why the provider omitted those records. Local-clock checks recover two 23-hour and two 25-hour days.

The audit checks 1,260 GEFS object receipts across 84 forecast cycles, members 1/10/20 and endpoints 24/30/36/42/48. Cached inputs contain all twenty common members. Initialization precedes issue by twelve hours; target leads from issue are 12-35 hours and from initialization 24-47 hours. Current object modification offsets do not authenticate historical first arrivals. Generic label interval-end clocks likewise do not represent source releases. Current revised electricity observations and the archive support delayed replay, not live-vintage certification.

Native GEFS resolution changes from one degree on 22 September 2020 to half a degree on 23 September. Integer-degree sampling fixes spatial support but retains that model-version boundary. The original quarterly marginal/R cutoff at origin minus seven days is only 6.5 days before the first issue; this boundary is retained when reproducing C3. Daily original calibration and new weekly conditional maps use actual issue minus seven days. An audit corrected new ν -selection cutoffs to actual first issue minus seven days and checked all sixteen selected choices: every ν and all eight factorial prediction columns remained identical. The final fixed $\nu=3$ selection excludes labels after 24 December 2023 12 UTC. New Danish fits enforce actual issue minus fifteen days throughout.

S9. Factorial intervention, marginal sensitivity and known-state experiments

S9.1. Fixed additional calibration and dependence choices

Forecast-only weather strata cross October-March/April-September with low wind/cold, other low wind and higher wind. The 2021 common-land-wind medians are 3.4972892 and 3.0565558 m/s; temperature lower quartiles are 2.3639913 and 10.7329014 degrees Celsius. These cutpoints are unchanged across periods and transferred regions.

Each Monday, additional recalibration fits a beta CDF to prior issued PITs in a 365-day stratum window ending seven days before the first issue. Contributing days receive equal weight in log-PIT likelihood. The fixed penalty four shrinks log shape parameters toward zero. Fewer than 28 represented dates or 168 hours triggers identity mapping. Parameters are fitted from permitted labels, not chosen to improve reanalysis scores. The old C3 Gaussian R is identical across the four factors. ν candidates are 3, 5, 8, 15, 30 and infinity; prior q90 event loss averages both margin paths. The first development quarter defaults to Gaussian, 2022 Q4 selects $\nu=8$, other development quarters select Gaussian, and the fixed reanalysis uses $\nu=3$. q95 inherits the primary choice.

The additional pooled q90 PIT coverage changes from 0.902069/0.885460/0.895345 to 0.905000/0.897816/0.900287 for DE-LU/FR/BE. At q95 it changes from 0.950000/0.930747/0.942759 to 0.953333/0.940345/0.947241. Pooled improvement in two areas is not a conditional-calibration certificate.

Table S9.1. Exact completed factorial scores.

Table S9.1b. Previously issued simple/reference forecasts read back on exactly the same factorial scoring domain. No new model fits or selection occurred. The original dynamic Gaussian is reproduced to below 2×10^{-12} in every period/threshold. The complete eleven-method comparison is retained in S4.

Table S9.2. Paired 14-calendar-day reanalysis intervals. Differences are candidate minus reference loss. Calendar positions with unavailable complete days remain zero-count positions, rather than being compressed away. All model losses are resampled synchronously within year/quarter, with 2,000 draws. Seven- and twenty-eight-day counterparts accompany the result records.

Period	Threshold	Model	Complete days	Event hours	Brier
development	q90	M0	715	675	0.029064210
development	q90	M1	715	675	0.028552284
development	q90	M2	715	675	0.029070923
development	q90	M3	715	675	0.028557321
development	q95	M0	715	308	0.014398478
development	q95	M1	715	308	0.014181040
development	q95	M2	715	308	0.014402240
development	q95	M3	715	308	0.014185469
reanalysis	q90	M0	725	2105	0.060190612
reanalysis	q90	M1	725	2105	0.060597924
reanalysis	q90	M2	725	2105	0.060186141
reanalysis	q90	M3	725	2105	0.060578659
reanalysis	q95	M0	725	1001	0.037109448
reanalysis	q95	M1	725	1001	0.037155451
reanalysis	q95	M2	725	1001	0.037072008
reanalysis	q95	M3	725	1001	0.037110642

Period	Threshold	Reference	Complete days	Brier
development	q90	Independence	715	0.028965243
development	q90	Static Gaussian	715	0.029239035
development	q90	Direct logistic	715	0.040411272
development	q90	Event climatology	715	0.045993862
development	q95	Independence	715	0.014072954
development	q95	Static Gaussian	715	0.014419423
development	q95	Direct logistic	715	0.019810464
development	q95	Event climatology	715	0.020419805
reanalysis	q90	Independence	725	0.061083921
reanalysis	q90	Static Gaussian	725	0.060207273
reanalysis	q90	Direct logistic	725	0.071667514
reanalysis	q90	Event climatology	725	0.111399388
reanalysis	q95	Independence	725	0.037213114
reanalysis	q95	Static Gaussian	725	0.037156215
reanalysis	q95	Direct logistic	725	0.045014430
reanalysis	q95	Event climatology	725	0.055556609

Threshold	Contrast	Loss difference	95% interval
q90	margin_at_G	+0.000407313	[-0.000356370, +0.001172379]
q90	dependence_at_base	-0.000004471	[-0.000093448, +0.000095466]
q90	margin_at_selected_dependence	+0.000392518	[-0.000369100, +0.001150914]
q90	dependence_at_calibrated	-0.000019265	[-0.000100497, +0.000069007]
q90	interaction	-0.000014794	[-0.000053602, +0.000016609]
q90	combined	+0.000388048	[-0.000407152, +0.001199289]
q95	margin_at_G	+0.000046004	[-0.000385836, +0.000464357]
q95	dependence_at_base	-0.000037439	[-0.000095011, +0.000016876]
q95	margin_at_selected_dependence	+0.000038633	[-0.000395321, +0.000457475]
q95	dependence_at_calibrated	-0.000044809	[-0.000101116, +0.000005122]
q95	interaction	-0.000007370	[-0.000031747, +0.000014449]
q95	combined	+0.000001194	[-0.000438482, +0.000425190]

Table S9.2b. Secondary relative Brier gain, $100(\text{BS_M1}-\text{BS_M3})/\text{BS_M1}$, in percent. Each draw resamples the denominator and paired difference from the same calendar blocks as Table S9.2. Positive values favour M3. The interpretation is precision for the prescribed finite-threshold forecast comparison; neither pointwise calibration certification nor a universal copula-equivalence claim is made.

Threshold	Block days	Relative gain (%)	95% interval (%)
q90	7	+0.03179	[-0.10673, +0.16987]
q90	14	+0.03179	[-0.10830, +0.17492]
q90	28	+0.03179	[-0.09603, +0.16659]
q95	7	+0.12060	[-0.02876, +0.26754]
q95	14	+0.12060	[-0.01354, +0.26608]
q95	28	+0.12060	[-0.01788, +0.26779]

S9.2. What the real sensitivity screen estimates

For fixed copula C , a common-uniform coupling bounds the event-probability change by $\min(1, \text{sum of absolute threshold-CDF changes})$. This elementary union bound is not novel. The cell residual mean $E\{1(Y \leq h) - v \mid \text{cell}\}$ estimates an averaged threshold discrepancy. Its absolute value can lower-bound mean absolute conditional error, but cannot upper-bound that error or certify each hourly probability.

The real stress path assumes an unknown constant CDF offset within a cell. A synchronous 14-calendar-day moving-block bootstrap supplies 2,000 joint residual draws within year/quarter. A maximum-standardised envelope covers six cells and ten residual dimensions. Monotone endpoint copula evaluations yield probability ranges; extrema of squared error then give conservative outer dependence-gain ranges. This is an assumed-path analysis, not an identified confidence set for every hourly conditional CDF. Positive endpoint movement or a fitted map is not a successful calibration claim.

Table S9.3. q90 reanalysis support and CDF residuals, original and additionally mapped margins. Zone order is DE-LU/FR/BE; gain is Gaussian loss minus selected-t loss under the mapped-margin stress path. Process membership overlaps across weather cells. Weather runs are contiguous hourly stretches, computed with explicit nanosecond normalization of pandas indices.

Weather cell	Days touched	Hours	Weather runs	Original residuals	Mapped residuals	Gain outer interval
cool_higher_wind	248	4421	126	-0.0113/+0.0066/+0.0055	-0.0093/+0.0122/+0.0149	[-0.0045617, +0.0045462]
cool_low_wind_cold	84	931	76	-0.1133/-0.0282/+0.0520	-0.1213/-0.0104/-0.0122	[-0.2331333, +0.2333393]
cool_low_wind_other	226	3264	159	-0.0599/+0.0366/-0.0049	-0.0373/+0.0238/-0.0073	[-0.0710531, +0.0708987]
warm_higher_wind	313	4709	229	+0.0091/+0.0290/+0.0204	+0.0066/+0.0475/+0.0211	[-0.0055376, +0.0054795]
warm_low_wind_cold	100	656	85	+0.0431/+0.0322/+0.0381	+0.0401/+0.0331/+0.0019	[-0.1010900, +0.1009028]
warm_low_wind_other	264	3419	239	-0.0025/+0.0545/+0.0121	-0.0066/+0.0494/+0.0091	[-0.0559896, +0.0561409]

All six gain intervals contain zero. They do not establish whether a true dependence contribution is absent or whether marginal error caused a specific increment. Remaining positive French/Belgian warm higher-wind residuals identify overstatement of average threshold exceedance risk. Small cold-cell process support limits precision. The full output includes both thresholds, all residual dimensions, observed event rates, conditional threshold positions and process support. A known-vector mapping check uses 35,050 finite prediction rows, including some incomplete-label days; that diagnostic count is not the 17,400-hour score sample.

S9.3. Explicit synthetic mechanisms and validation sequence

All new simulated observations are synthetic. Observed state X equals minus or plus one with probability one half; normal marginal means are X times $(0.4, 0.3, 0.2)$, with unit variance and threshold Φ -inverse(0.9). Conditional correlation uses the positive-definite partial-correlation transform in S3 with $\eta = (0.55, 0.45, 0.30) + X(0.25, 0.20, 0.15)$. Correct-Gaussian and true-t5 cases share those margins and state correlation. Marginal misspecification shifts issued means by 0.25 and scales standard deviation by 0.8. The hidden-weather case mixes equally weighted Gaussian correlations at η plus/minus $(0.65, 0.65, 0.65)$, conditional on observed X ; the Gaussian comparator uses their exact mixture covariance. Thus its observed-state distribution is already non-Gaussian; no physical tail mechanism is inferred.

Each repetition generates separate prior calibration, candidate-selection and new-evaluation blocks. Six state-specific threshold-CDF intervals and two event-frequency intervals use Bonferroni Clopper-Pearson coverage, at least 95% familywise under the IID DGP. Smoothed threshold estimates are $(\text{count} + 0.5) / (\text{state count} + 1)$. ν selection minimises equally weighted loss over original and mapped margins, with fixed simplicity preference for exact ties. Gaussian ranges must be disjoint from the event interval and a finite-t range intersect it in at least one observed state. The final flag also requires a positive independent paired Brier-gain 95% t interval. This latter calculation exploits synthetic IID sampling, not real weather-hour independence.

An exploratory compatibility-only run exposed ineffective flags under hidden weather. The score requirement was added during development; final independent-seed runs then validated that developed rule. They are not preregistered experiments. The high-information run uses 100,000 calibration, 10,000 selection and 100,000 new-test rows per repetition, with 100 repetitions per mechanism. The lower-information run uses 10,000 in each block; selection size is unchanged. Known correlations deliberately remove R-estimation uncertainty.

Table S9.4. Known-mechanism simulations, 100 independent repetitions per mechanism. 'Compatibility' concerns finite-threshold probabilities after marginal uncertainty. The final flag additionally requires positive independent paired-score evidence.

Synthetic mechanism	Finite selected ν	Compatibility alone	Final flag	95% flag-rate interval
Correct Gaussian	34/100	0/100	0/100	[0.0%, 3.6%]
Gaussian with marginal error	27/100	0/100	0/100	[0.0%, 3.6%]
True t5	99/100	70/100	70/100	[60.0%, 78.8%]
Hidden Gaussian-weather mixture	54/100	36/100	0/100	[0.0%, 3.6%]

True-t5 mean new-test paired gain in the high-information run is 0.0000288339866, with repetition-level Monte Carlo interval $[0.0000269481088, 0.0000307198644]$. The hidden-weather mixture has mean gain -0.0000014817984 $[-0.0000026080521, -0.0000003555446]$; all 36 compatibility flags fail independent benefit verification. No flags in 100 repetitions yields upper 95% rate bound 0.0362167, not a guaranteed zero-error rule. The lower-information t5 mean gain remains positive, 0.0000303957370 $[0.0000248946592, 0.0000358968148]$, but no repetition passes the final screen. Large homogeneous IID counts cannot be substituted for the effective information in persistent regional weather processes.

S9.5. Joint reliability of the factorial probabilities

Reliability uses the probability boundaries fixed in S4, without bin selection from reanalysis outcomes. All four forecast streams and both thresholds share 725 complete UTC days (17,400 hours). Daily bin counts, events and probability sums are resampled synchronously in 2,000 draws of 14-calendar-day blocks within year-quarter strata. The full 731-day calendar retains unavailable dates; sampled stratum contributions retain the original valid-day weighting. Intervals describe observed frequency minus mean issued probability and are pointwise percentile intervals. Empty bins remain visible; an interval is withheld if fewer than 95% of draws contain observations in that bin.

For q90, every bin interval includes zero for M0, M1 and M3; M2 has a small negative lowest-bin gap based on two event days. For q95, the [0.6,0.8) bin retains overprediction after additional marginal mapping: M1 predicts 0.689068 against frequency 0.569811 across 265 hours and 54 event days, whereas M3 predicts 0.688513 against 0.578755 across 273 hours and 58 event days. Their signed-gap intervals are $[-0.211300, -0.012593]$ and $[-0.199996, -0.008891]$. Neither modification removes every probability departure. Hours and event days measure support rather than independent replications. Intervals spanning zero do not certify conditional calibration; intervals excluding zero are descriptive, without simultaneous coverage across bins or evidence that model-specific gaps differ.

Table S9.5. Original-region joint reliability on common 2024-2025 scoring days. Probabilities and signed-gap intervals are on the 0-1 scale. Event days count distinct UTC dates containing an event within that forecast bin. NA denotes an empty or insufficiently represented interval; no bins are deleted.

Original-region fixed-bin joint reliability by threshold and M0-M3: supporting-table file **supporting_table_01.csv** (80 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

S10. Forecast-only allocation, exact exposure matching and process uncertainty

The frozen primary forecast is C3/M0 q90; q95 applies the same gates. P1 gate 0.25 and P2 gate 0.19 are obtained on a fixed grid using 2022 workload. A six-hour cooldown is selected using 2023 early process coverage against P1, among six, twelve and twenty-four hours. The earlier C3 specification had already used the full development period: this is chronological strategy development, not fully nested validation. No reanalysis or Danish outcome tunes a rule. The score $p_t[1-\max \text{ previous six issued risks}]$ is a priority, not onset probability.

Rules allocate on complete forecast windows, before any future-outcome completeness restriction. Cooldown and past risk carry across outcome-missing days and evaluation boundaries. The score domain is subsequently restricted to common complete-label UTC days. In the original-region q90 reanalysis, planned alerts on all forecast-complete days are 1,454/619/573 for P0/P1/P2; the complete-label counts are 1,450/618/572. q95 all-plan counts are 1,454/405/371, versus 1,450/405/371 in scoring. Missing future labels cannot reset priority or cooldown. Synthetic truth-flip/missing-label checks leave plans unchanged; future forecast changes leave earlier daily plans unchanged.

Matching retains exactly B slots uniformly from each N-slot plan, without replacement or event-dependent choice. Event-hour detection expectation is original detected count times B/N. Process detection is $1-\text{choose}(N-m,B)/\text{choose}(N,B)$, integrating randomisation exactly. These are retrospective evaluation expectations conditional on fixed plans, not a deployable controller knowing its annual candidate count. Non-integer matched detection is expected randomised detection. Daily maximum two is preserved. Zero-alert-day counts refer to native plans before thinning.

Table S10.1. Native and exact matched-workload results with one inactive-day bridge.

Threshold	Policy	Mode	Slots	Expected event-hour hits	First-six-hour coverage	Any-process coverage	Native zero-alert days
q90	P0_forced_top2	native	1450	341.000	0.49296	0.91549	0
q90	P0_forced_top2	matched	572	134.519	0.24833	0.71249	0
q90	P1_risk_gate	native	618	322.000	0.45070	0.88732	401
q90	P1_risk_gate	matched	572	298.032	0.43174	0.87565	401
q90	P2_risk_rise_cooldown	native	572	259.000	0.47887	0.88732	371
q90	P2_risk_rise_cooldown	matched	572	259.000	0.47887	0.88732	371
q95	P0_forced_top2	native	1450	188.000	0.47541	0.86885	0
q95	P0_forced_top2	matched	371	48.102	0.16537	0.48035	0
q95	P1_risk_gate	native	405	166.000	0.40984	0.80328	510
q95	P1_risk_gate	matched	371	152.064	0.39186	0.78262	510
q95	P2_risk_rise_cooldown	native	371	115.000	0.44262	0.68852	478
q95	P2_risk_rise_cooldown	matched	371	115.000	0.44262	0.68852	478

Table S10.2. P2 first-six-hour coverage minus comparator, percentage points. Intact realised process contributions are attributed to onset days and synchronously resampled in calendar blocks within year/quarter. This preserves spell endpoints and does not create new processes at concatenated bootstrap boundaries. Intervals condition on policies, retention fractions and realised spell boundaries; policy-selection and full process-definition uncertainty are not included. Zero-/two-inactive-day bridges are reported in the accompanying process records.

Threshold	Comparator	Block days	Difference (pp)	95% interval (pp)
q90	P0_forced_top2	7	+23.054	[+12.794, +35.097]
q90	P1_risk_gate	7	+4.714	[-5.858, +17.841]
q90	P0_forced_top2	14	+23.054	[+15.119, +37.920]
q90	P1_risk_gate	14	+4.714	[-4.704, +19.723]
q90	P0_forced_top2	28	+23.054	[+18.110, +39.483]
q90	P1_risk_gate	28	+4.714	[-2.621, +20.094]
q95	P0_forced_top2	7	+27.726	[+14.619, +39.159]
q95	P1_risk_gate	7	+5.076	[-10.994, +19.535]
q95	P0_forced_top2	14	+27.726	[+15.441, +38.506]
q95	P1_risk_gate	14	+5.076	[-10.560, +18.570]
q95	P0_forced_top2	28	+27.726	[+14.927, +37.504]
q95	P1_risk_gate	28	+5.076	[-10.670, +17.070]

Table S10.2b. Direct P1-P0 early-coverage contrast, at exactly the same common budgets as Table S10.1. The same process-contribution bootstrap is used; uncertainty intervals are computed directly, not by subtracting endpoints from two other contrasts.

Threshold	Block days	Difference (pp)	95% interval (pp)
q90	7	+18.341	[+11.866, +24.390]
q90	14	+18.341	[+13.086, +24.862]
q90	28	+18.341	[+15.461, +25.444]
q95	7	+22.649	[+15.066, +29.976]
q95	14	+22.649	[+15.996, +30.812]
q95	28	+22.649	[+16.741, +28.946]

Table S10.3. Descriptive yearly native workload and coverage. Rules remain fixed. Processes are assigned by onset year while their full recorded boundaries remain intact across January 1; target-hour slots and hits are attributed by target year. These are secondary summaries, not new model-selection samples.

Threshold	Year	Policy	Slots	Event-hour hits	First-six-hour processes
q90	2024	P0_forced_top2	726	216	18/39
q90	2024	P1_risk_gate	356	203	15/39
q90	2024	P2_risk_rise_cooldown	334	169	23/39
q90	2025	P0_forced_top2	724	125	17/32
q90	2025	P1_risk_gate	262	119	17/32
q90	2025	P2_risk_rise_cooldown	238	90	11/32
q95	2024	P0_forced_top2	726	122	14/38
q95	2024	P1_risk_gate	254	114	13/38
q95	2024	P2_risk_rise_cooldown	233	80	16/38
q95	2025	P0_forced_top2	724	66	15/23
q95	2025	P1_risk_gate	151	52	12/23
q95	2025	P2_risk_rise_cooldown	138	35	11/23

Figure S1. Post hoc examples of gains and losses from risk-rise priority. The same two previously selected cases illustrate directional differences between P1 and P2. Upper panels show issued joint risk; lower lanes separate forecast-only planned hours and realised event hours. The tan bands mark the first six elapsed hours after observed onset. The grey P2 band in the second case marks the cooldown after a planned hour at -1 h. Time is relative to 00 UTC on the date shown. Observed onset and events annotate the cases and do not trigger allocation. These illustrations are not independent predictive validation.

S10.4. Dependence changes under the same fixed gate

The P1 probability gate 0.25 and daily maximum two are applied without retuning to all four forecast streams. Forecast-only plans are generated over the available full history before the scoring window is taken. All models are scored on common complete-label days. Exposure matching uses the smallest native slot count across the four streams for each threshold. Paired M3-M1 intervals retain intact process contributions and use onset-day calendar blocks. These are secondary model-to-decision checks, conditional on fixed plans and retention fractions.

Table S10.4. Native fixed-P1 results in the original regions.

Threshold	Forecast	Native slots	Selected event hours	Early processes
q90	M0	618	322	32/71
q90	M1	619	327	31/71
q90	M2	616	320	31/71
q90	M3	613	326	31/71
q95	M0	405	166	25/61
q95	M1	405	169	25/61
q95	M2	404	167	25/61
q95	M3	406	170	25/61

Table S10.5. M3-M1 expected first-six-hour coverage at common workload, percentage points.

Threshold	Common slots	Block days	Difference (pp)	95% interval (pp)
q90	613	14	+0.2201	[-1.2407, +4.6556]
q90	613	7	+0.2201	[-1.3375, +4.6823]
q90	613	28	+0.2201	[+0.1439, +4.7965]
q95	404	14	-0.0486	[-0.0724, -0.0252]
q95	404	7	-0.0486	[-0.0722, -0.0276]
q95	404	28	-0.0486	[-0.0647, -0.0204]

At q90 the M3 and M1 plans differ: eleven selections are added and seventeen removed. One early process is gained and another lost, leaving both at 31/71. The 14-day interval contains zero whereas the 28-day interval is positive; all block results are retained. At q95 all four native early counts equal 25/61. Its small negative matched M3-M1 difference arises from the differing retention fractions, not a new loss of native early processes. A narrow conditional interval for that randomisation effect should not be read as a substantial operating difference. The Danish counterparts use the same rules and are reported in S11.

S10.6. Fixed plans under alternative French label timestamps

The S7 label-only sensitivity is extended to the new review rules without changing forecasts, thresholds or allocation. We reuse the existing S7 alternative-event tables, constructed by shifting native French load, wind and solar records jointly by -30 or $+30$ minutes before complete-hour aggregation. German and Belgian

observations remain unchanged. Both alternatives retain the original physical thresholds. Their original S7 manifests bind the generated labels to their source files; this extension verifies those hashes, verifies unchanged C3 probability bits, and reads the existing v3 P0/P1/P2 plans generated before outcome scoring. Prior-history allocation state is therefore retained. The original convention remains the primary definition; neither alternative is selected by its results.

Each convention is first evaluated on its own complete-label scoring days. A second evaluation uses the 722 days jointly complete under all three conventions, separating label changes from differences in eligibility. In every evaluation, all policies share $B=\min(N_{P1},N_{P2})$ for exact exposure standardisation; both P1-P0 and P2-P1 use that same B. Active-day spells retain the one-observed-inactive-day bridge. First-six-hour intervals use 2,000 paired 14-calendar-day year-quarter onset-day block draws of intact spell contributions. They condition on the fixed plans, thinning fractions and convention-specific realised boundaries, without treating outcome-defined spells as independent weather episodes.

Original-convention results on its 725-day domain reproduce the main v3 results exactly. On the common 722-day q90 domain, P1-P0 matched early-coverage gains are 18.25, 17.02 and 17.83 percentage points for original, -30-minute and +30-minute labels, with all three pointwise intervals above zero. P2-P1 intervals include zero under every convention. Under -30-minute labels, native P2 covers 30/75 spells early, below P1's 31/75; its small positive matched difference does not imply native superiority. The q95 results also retain a positive matched P1-P0 contrast but not a resolved P2-P1 increment. These are fixed-forecast measurement-convention checks, not certification of a French timestamp interpretation or robustness of a fully refitted pipeline.

Table S10.6a. Each convention's own complete-label scoring domain. Native early counts use the number of realised active-day spells as denominator. Matched quantities are analytic uniform-thinning expectations, not an additional issued policy.

Threshold	Label convention	Complete days	Policy	Native slots	Native detected event hours	Native early spells	Common B	Expected matched event hours	Expected matched early coverage (%)
q90	Original	725	P0	1450	341	35/71	572	134.519	24.833
q90	Original	725	P1	618	322	32/71	572	298.032	43.174
q90	Original	725	P2	572	259	34/71	572	259.000	47.887
q90	RTE -30 min	722	P0	1444	345	34/75	571	136.423	22.392
q90	RTE -30 min	722	P1	618	327	31/75	571	302.131	39.410
q90	RTE -30 min	722	P2	571	262	30/75	571	262.000	40.000
q90	RTE +30 min	725	P0	1450	343	36/71	572	135.308	25.052
q90	RTE +30 min	725	P1	618	324	32/71	572	299.883	42.979
q90	RTE +30 min	725	P2	572	263	33/71	572	263.000	46.479
q95	Original	725	P0	1450	188	29/61	371	48.102	16.537
q95	Original	725	P1	405	166	25/61	371	152.064	39.186
q95	Original	725	P2	371	115	27/61	371	115.000	44.262
q95	RTE -30 min	722	P0	1444	203	28/62	371	52.156	16.225
q95	RTE -30 min	722	P1	405	177	25/62	371	162.141	38.678
q95	RTE -30 min	722	P2	371	126	27/62	371	126.000	43.548
q95	RTE +30 min	725	P0	1450	182	29/59	371	46.567	16.128
q95	RTE +30 min	725	P1	405	159	24/59	371	145.652	38.700
q95	RTE +30 min	725	P2	371	112	27/59	371	112.000	45.763

Table S10.6b. Common 722-day domain under all three label conventions. Native early counts use the number of realised active-day spells as denominator. Matched quantities are analytic uniform-thinning expectations, not an additional issued policy.

Threshold	Label convention	Complete days	Policy	Native slots	Native detected event hours	Native early spells	Common B	Expected matched event hours	Expected matched early coverage (%)
q90	Original	722	P0	1444	341	35/71	571	134.841	24.884
q90	Original	722	P1	618	322	32/71	571	297.511	43.130
q90	Original	722	P2	571	259	34/71	571	259.000	47.887
q90	RTE –30 min	722	P0	1444	345	34/75	571	136.423	22.392
q90	RTE –30 min	722	P1	618	327	31/75	571	302.131	39.410
q90	RTE –30 min	722	P2	571	262	30/75	571	262.000	40.000
q90	RTE +30 min	722	P0	1444	343	36/71	571	135.632	25.104
q90	RTE +30 min	722	P1	618	324	32/71	571	299.359	42.931
q90	RTE +30 min	722	P2	571	263	33/71	571	263.000	46.479
q95	Original	722	P0	1444	188	29/61	371	48.302	16.599
q95	Original	722	P1	405	166	25/61	371	152.064	39.186
q95	Original	722	P2	371	115	27/61	371	115.000	44.262
q95	RTE –30 min	722	P0	1444	203	28/62	371	52.156	16.225
q95	RTE –30 min	722	P1	405	177	25/62	371	162.141	38.678
q95	RTE –30 min	722	P2	371	126	27/62	371	126.000	43.548
q95	RTE +30 min	722	P0	1444	182	29/59	371	46.760	16.190
q95	RTE +30 min	722	P1	405	159	24/59	371	145.652	38.700
q95	RTE +30 min	722	P2	371	112	27/59	371	112.000	45.763

Table S10.6c. Paired early-coverage contrasts under the shared three-policy budget. The original full-domain rows retain the main 572-slot q90 and 371-slot q95 comparison; the common q90 domain uses 571 slots. Intervals are descriptive pointwise comparisons.

French timestamp alternatives: paired early-coverage contrasts: supporting-table file **supporting_table_02.csv** (24 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

S11. Danish specification transfer, measurement and weather support

The transfer combinations are Germany-Luxembourg/France/DK1 and Germany-Luxembourg/France/DK2. Energinet settlement data supply complete 2019-2025 hourly records with explicit bidding-area identifiers. Following the provider accounting warning, load excludes uncertain commercial self-consumption before subtracting all reported wind and PV categories; the residual equals through-grid consumption minus grid-injected PV and wind. Signed generation and negative net load remain in the sample. Current metadata specifies 9-15-day initial publication and revisions up to two years (Energinet, 2026).

New Danish thresholds, marginal models and three-region correlations use only observation hours ending at least fifteen elapsed days before the relevant first issue. Danish load lags are 21 and 28 days and follow the same hour-end availability rule. The original German-Luxembourg and French issued margins are reused with their recorded clock; the fifteen-day rule is not claimed for those cached fits. Local GEFS summaries replace the former Belgian weather slot for the Danish margin, and the joint weather basis averages the three relevant regions. Fixed approximate land polygons and marine windows use common integer-degree nodes, twenty members and the original hourly interpolation (Natural Earth, n.d.). The macro-weather-only version is an input sensitivity.

Model settings, $\nu=3$, numerical cutpoints and warning rules were transferred unchanged. Local weather was declared primary before recorded Danish performance inspection; its geographic extraction and input schema were finalised independently of electricity labels before scoring the primary local profile. Only region-specific numerical parameters are fitted on permitted history: this is specification transfer, not zero-shot transfer of fitted correlations. We also evaluate the Danish-pivotal component, requiring Danish exceedance and exactly one original-region exceedance, to distinguish new information from the reused Germany-Luxembourg/French pair. S11 records accounting, masks, clock checks and secondary outcomes.

Energinet PriceArea explicitly distinguishes DK1 west and DK2 east of the Great Belt. The 2019-2025 extraction contains 61,368 records per zone and no explicit core-component nulls or duplicate zone/UTC identifiers. Request filters follow the provider's local-time API specification; padded intervals are trimmed to exact UTC years. Every supplied local timestamp agrees with UTC conversion. Source credit and transformed-field definitions follow the provider's CC BY 4.0 conditions.

Let G denote GrossConsumptionMWh, A commercial/local self-consumption, S_{self} estimated solar self-consumption, S_{grid} the sum of the other three PV classes, and W all four wind classes. Corrected total load is $G-A$ and reported PV is $S_{\text{self}}+S_{\text{grid}}$. Net load is therefore $G-A-S_{\text{self}}-S_{\text{grid}}-W$, equal to provider through-grid consumption minus grid PV and wind. This avoids subtracting solar self-consumption twice. One-hour MWh converts to mean MW. Domestic grid losses remain included under the source convention. This is not harmonised gross load across all countries.

In 2019 all PV was grouped in the self-consumption category; commercial/local separation changes from 2021. Zero category entries do not imply physical absence. Gross and corrected load are positive throughout. Tiny negative total PV appears in 1,251 hours in 2020 (683 DK1 and 568 DK2), with minima -0.6 and -0.5 MW, and is retained without clipping. There are 15,510 negative net-load hours; renewable surplus can be

balanced by export. Provider warnings about some zero-as-null values remain unresolved rather than being silently imputed. No score-dependent row removal is used.

A pretest 2019-2020 SCADA comparison contains gross load anomalies, including DK1 964,623.8 and 321,545.7 MW on 27-28 February 2020. Together with source warnings about unrepaired SCADA errors, these justify the settlement choice before joint scoring. This diagnosis does not certify every settlement observation. Revised settlement snapshots, first-publication timing and later revision horizons remain separate issues.

Local land masks derive from the validated Natural Earth 1:10m country polygons, version 5.1.1, clipped to 4-17°E and 53-59°N. Components with representative longitude below 11.1°E or latitude above 56.6°N are assigned DK1; others DK2. This approximates Danish island groups, including a northern-island rule for Læsø/Anholt, rather than recovering an official electrical polygon. Land areas are approximately 32,990.17 and 9,705.85 square km. DK1 marine support is 5-8.8°E,54-57.5°N and DK2 support 11-16°E,54-56.5°N, subtracting Natural Earth land; areas are 85,840.49 and 56,204.28 square km. They are geographic weather proxies, not EEZs or installed-capacity weights.

Integer-degree nodes represent one-degree cell footprints. Intersections with masks are measured in equal-area EPSG3035 and normalised as area weights. The same nodes are sampled from later half-degree native grids. All twenty perturbed members and five original endpoints are preserved. u/v components are interpolated at hourly midpoints before node wind norms and regional averaging. Solar redistribution conserves the original six-hour mean. Ensemble summaries use member means and sample standard deviations. The old rectangular-mask geographic pilot is not used as a primary modelling input.

S11. Regional transfer with prespecified local weather

The complement of the Danish-pivotal subset is the set of hours when Germany-Luxembourg and France both exceed, irrespective of whether Denmark also exceeds. It contains the same 924 q90 hours in both combinations and can include three-area exceedances; it is not a category in which only the two reused areas exceed.

The main extension uses Danish local ensemble weather; unchanged macroregional weather is a prespecified input sensitivity. Both Danish combinations are reported irrespective of score ordering. The German-French components overlap the original analysis, and DK-pivotal results are nested secondary diagnostics rather than independent replications. All reported intervals are pointwise. No threshold levels, model degrees of freedom or warning parameters were reselected in Denmark; the physical thresholds themselves were fitted from prior Danish observations under the transferred specification.

Table S11.1. Complete model loss comparison for q90 and q95. All model scores within each combination use the same complete-day domain. Independence uses the base marginal CDFs. Score differences across weather inputs are input sensitivity results and do not determine the primary version.

Danish transfer: joint Brier scores by weather input, threshold and model: supporting-table file **supporting_table.03.csv** (20 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Table S11.2. Paired 14-day loss contrasts. Differences are candidate minus reference loss; negative values favour the candidate. The marginal contrasts are M1-M0 at Gaussian dependence and M3-M2 at t dependence. The dependence contrasts are M2-M0 with base margins and M3-M1 with conditional margins. Interaction is (M3-M2)-(M1-M0), and combined is M3-M0. The primary and DK-pivotal outcomes use synchronous elapsed-calendar blocks and the same realised missing-date positions. Seven- and 28-day results are retained in the accompanying complete contrast table; none is used to reselect a model.

Danish transfer: paired factorial Brier contrasts for joint and pivotal outcomes: supporting-table file **sup-**

porting_table_04.csv (96 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Table S11.3. Secondary fixed-P1 model sensitivity. Every model uses the probability gate 0.25, at most two slots per day and no cooldown. Counts in the native column are actual detections of outcome-defined active-day spells. Common B is the smallest native model workload within each combination and threshold. Matched values are exact uniform-thinning expectations. Small expected differences with identical native spell counts do not represent newly detected spells.

Danish transfer: fixed P1 native and matched early coverage across model streams: supporting-table file **supporting_table_05.csv** (32 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Table S11.4. Fixed-P1 M3-M1 coverage contrasts. Calendar blocks resample onset-day contributions of complete outcome-defined active-day spells within year and quarter, conditional on the fixed rules and retained fractions; they are not simultaneous confidence bands. All three block lengths are reported.

Danish transfer: fixed P1 M3-M1 coverage contrasts across block lengths: supporting-table file **supporting_table_06.csv** (24 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Table S11.5. Risk gating and fixed-priority contrasts at matched review exposure. P1-P0 evaluates whether permitting zero alerts improves early coverage relative to a daily forced quota. P2-P1 evaluates the additional fixed priority and cooldown rule. Comparisons use each combination's own common B and retain full forecast history before scoring. These exposure-standardised comparisons do not describe a prospective exact annual-quota controller.

Danish transfer: policy coverage contrasts across block lengths: supporting-table file **supporting_table_07.csv** (48 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Table S11.6. Year-specific native workload and coverage. Target-hour slots and hits are assigned by target year; outcome-defined active-day spells are assigned by first active hour's year while their complete recorded boundaries persist across January 1. This is a descriptive stability summary, not an additional selection sample.

Danish transfer: annual native allocations, hits, early spells and zero-alert days: supporting-table file **supporting_table_08.csv** (48 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Joint probability reliability. The fixed bins are [0,0.005,0.01,0.02,0.05,0.1,0.2,0.4,0.6,0.8,1]. The common domain requires all eight model probabilities and both event labels to be finite on a complete 24-hour day. Bootstrap sampling preserves the full 731-day calendar with missing-date contributions equal to zero, using 2000 paired 14-day blocks within year and quarter and the original reliability analysis's valid-day weights. Hours, days and event days report support rather than independent sample sizes. Empty bins retain NA; intervals are suppressed unless at least 95% of bootstrap repetitions contain observations in the bin. An interval containing zero is not proof of calibration. For DK1 q90, the fixed [0.2,0.4) probability bin contained 1233 hours on 257 days and 356 events on 117 days; its observed-minus-predicted gap was -0.0006, with pointwise 95% interval [-0.0426, +0.0736]. For DK1 q95, the fixed [0.2,0.4) probability bin contained 678 hours on 169 days and 192 events on 68 days; its observed-minus-predicted gap was +0.0006, with pointwise 95% interval [-0.0731, +0.0597]. For DK2 q90, the fixed [0.2,0.4) probability bin contained 1402 hours on 286 days and 405 events on 134 days; its observed-minus-predicted gap was +0.0035, with pointwise 95% interval [-0.0341, +0.0698]. For DK2 q95, the fixed [0.2,0.4) probability bin contained 796 hours on 190

days and 207 events on 84 days; its observed-minus-predicted gap was -0.0217, with pointwise 95% interval [-0.0821, +0.0406].

Table S11.7. Main local-weather M0 bin support and reliability. Gaps equal observed frequency minus issued mean probability; positive gaps indicate underprediction. The full M0-M3 local and macro bin results, support and sparse-bin flags are retained in the accompanying reliability table.

Danish local-weather transfer: fixed-bin joint reliability: supporting-table file **supporting_table_09.csv** (40 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

Information-time and input checks. All 2557 weather cycles, 20 members per cycle and 61,368 target hours were verified before the local models ran. The completed summary's SHA-256 matched both producer and consumer receipts; its UTC hourly index, finite physical fields and temperature units were checked. Temperature means arrive in kelvin and are converted to Celsius for prediction; spreads retain their numerical values. Danish quarterly marginal fitting, threshold fitting and dependence fitting, daily calibration, weekly conditional calibration and lag availability require at least 15 elapsed days after the observed interval ends. Danish consumption lags are 21 and 28 days. Current revised settlement observations do not authenticate historical publication vintages. Local weather was declared primary before recorded Danish performance inspection; its final geographic extraction and input schema preceded primary local scoring and were specified independently of electricity labels. The masks approximate Danish electrical areas geographically; their marine support is fixed rather than capacity weighted. The corresponding inherited German-French margins retain their original information specification.

S12. Limited multiplicity checks and sensitivity digest

Post hoc finite-family multiplicity check

The scientific primary comparisons remain additional t dependence at mapped margins (M3-M1) and onset-priority versus gating (P2-P1), both for q90. The gate-versus-quota result is a secondary operational comparison identified during the reanalysis, not a newly designated preregistered primary endpoint. This reporting check was added after the contrasts had been inspected. It does not make the reanalysis confirmatory.

Four finite interpretation families are defined: q90 model contrasts M1-M0 and M3-M1 in each of the original and two primary local-weather Danish combinations (six tests); the corresponding six q90 policy contrasts P1-P0 and P2-P1; and separate secondary q95 model and policy families with the same structure. Holm adjustment is applied within each six-test family. Macro-weather inputs, block-length changes, weather-cell diagnostics, French label timing, and alternative process bridging remain descriptive sensitivities; the adjustment does not cover every supplementary comparison or establish familywise control for the complete research search.

Each statistic uses the same paired 14-day calendar scheme and random draws as its existing pointwise percentile interval. The model scheme is synchronous circular elapsed-calendar resampling with missing positions retained. Policy process contributions are paired, fixed, and resampled by onset day within year-quarter strata; realised episodes are not reconnected. The two-sided null approximation compares $|T^*-T|$ with $|T|$ and uses $(1 + \text{number of exceedances}) / (1 + \text{number of defined draws})$, with 2,000 replicates. This centered-bootstrap approximation is neither an uncentered bootstrap sign fraction nor an exact randomisation test; finite-sample validity depends on the corresponding resampling approximation. Existing pointwise intervals are reproduced exactly and remain pointwise, not Holm-adjusted intervals. Shared German/French data mean the three regional comparisons are dependent; Holm does not require independence of valid input p-values.

model_q90

zone	display_contrast	estimate	pointwise_ci_low	pointwise_ci_high	raw_centered_ bootstrap _p	holm_adjusted_p
DE_LU_FR_BE	M1-M0	0.0004073125	-0.0003563696	0.001172379	0.2903548	1
DE_LU_FR_BE	M3-M1	-1.92649e-05	-0.0001004969	6.900721e-05	0.6576712	1
DK1	M1-M0	-0.0002426354	-0.0007856512	0.0002843586	0.3728136	1
DK1	M3-M1	2.222091e-05	-3.621692e-05	8.262187e-05	0.4477761	1
DK2	M1-M0	-0.0003504439	-0.001222508	0.0004766059	0.4097951	1
DK2	M3-M1	2.443634e-05	-6.564912e-05	0.0001096378	0.5787106	1

model_q95

zone	display_contrast	estimate	pointwise_ci_low	pointwise_ci_high	raw_centered_ bootstrap _p	holm_adjusted_p
DE_LU_FR_BE	M1-M0	4.600366e-05	-0.0003858357	0.000464357	0.84008	1
DE_LU_FR_BE	M3-M1	-4.480939e-05	-0.0001011161	5.122185e-06	0.09995002	0.5997001
DK1	M1-M0	-0.0002181084	-0.0005644217	0.000117639	0.2093953	0.8375812
DK1	M3-M1	5.252385e-05	-1.301011e-05	0.0001201296	0.1274363	0.6371814
DK2	M1-M0	-0.0001458095	-0.0007520976	0.0003090478	0.5977011	1
DK2	M3-M1	4.220848e-05	-2.860361e-05	0.0001102267	0.2313843	0.8375812

policy_q90

zone	display_contrast	estimate	pointwise_ci_low	pointwise_ci_high	raw_centered_ bootstrap _p	holm_adjusted_p
DE_LU_FR_BE	P1-P0	0.1834057	0.1308597	0.2486212	0.0004997501	0.002998501
DE_LU_FR_BE	P2-P1	0.04713807	-0.04704442	0.1972311	0.5112444	1
DK1	P1-P0	0.227781	0.160503	0.2757555	0.0004997501	0.002998501
DK1	P2-P1	0.03117904	-0.05684902	0.1521804	0.5777111	1
DK2	P1-P0	0.1796334	0.1162004	0.2354419	0.0004997501	0.002998501
DK2	P2-P1	0.04833676	-0.07041721	0.1835077	0.4757621	1

policy_q95

zone	display_contrast	estimate	pointwise_ci_low	pointwise_ci_high	raw_centered_ bootstrap _p	holm_adjusted_p
DE_LU_FR_BE	P1-P0	0.2264927	0.1599598	0.3081161	0.0004997501	0.002998501
DE_LU_FR_BE	P2-P1	0.05076316	-0.1056012	0.1857048	0.4907546	1
DK1	P1-P0	0.2218796	0.1065014	0.2705049	0.0004997501	0.002998501
DK1	P2-P1	0.04457957	-0.07759779	0.2089612	0.5542229	1
DK2	P1-P0	0.2833481	0.2042941	0.3619705	0.0004997501	0.002998501
DK2	P2-P1	-0.03478764	-0.1594055	0.09994427	0.5977011	1

Interpretation

All three q90 gate-versus-quota comparisons remain supported in this post hoc check (largest Holm-adjusted $p=0.002999$). None of the q90 added-dependence or onset-priority contrasts rejects zero after the respective finite-family adjustment. This strengthens the reporting discipline for the gating result while preserving uncertainty about additional t dependence and priority logic. It does not establish model equivalence, absence of tail dependence, prospectively verified dispatch value, or multiplicity control for all analyses.

Robustness digest for the main interpretation

The q90 probability-gate advantage remains positive in all three primary combinations under 0-, 1-, and 2-empty-day process bridging (descriptive matched-coverage gains span 17.96-23.62 percentage points). At q95 it is also positive for these bridging definitions. The gate-versus-quota pointwise intervals exclude zero at each 7-, 14-, and 28-day block length in both thresholds and all three primary combinations. The French ± 30 -minute label alternatives retain a q90 gate gain of 17.02-18.25 percentage points on their common 722-day scoring domain, with intervals above zero. These are fixed-plan label checks, not refitted alternative historical pipelines.

The primary local-weather Danish q90 gains are 22.78 and 17.96 percentage points; the macro-input sensitivity gives 17.32 and 12.73, with their 14-day intervals above zero. Macro weather was not selected by performance. Added t dependence and the risk-rise/cooldown rule do not show a stable incremental advantage; the original P2-P1 native early detection difference reverses from +8 processes in 2024 to -6 in 2025. Bridging, block, threshold, and input sensitivities cannot remove this warning about priority transfer.

The P1 threshold 0.25, maximum two daily slots, P2 threshold 0.19, and six-hour cooldown are fixed transferred settings. No new neighbourhood sweep of those parameters was run in this revision. Their fixed transfer across thresholds, years, and two Danish combinations is evidence about that specification, not evidence that nearby gates, different daily caps, or all cooldowns are equally effective. q90 defines relatively frequent elevated residual-demand episodes; q95 is the secondary rarer-event check. Neither percentile is an empirically validated adequacy or outage threshold. All sensitivity comparisons are descriptive and are not added to the finite Holm families.

S13. Operational interface, market context and accounting scale

S13. Operational context and exploratory market association

S13.1. Intended review interface and information timing

The allocation experiment selects delivery-hour windows for review. A proposed implementation would open a case containing the forecast area probabilities, joint probability, weather-cycle identity and issue time, then check available generation and reserves, support available from neighbouring areas, exchange capacity and network outages. This check sequence is compatible with the demand, generation and cross-border information used in ENTSO-E's daily short-term adequacy assessment and regional coordination services. The study did not use confidential TSO adequacy, reserve or security assessment records and did not route forecasts into those services (ENTSO-E, n.d., OPC/STA; Regions).

For target day D, the archived GEFS cycle is initialised at D-1 00 UTC and forecasts are issued at D-1 12 UTC for D 00-23 UTC, giving 12-35 h delivery lead. The normal day-ahead auction closes around noon local market time. Nord Pool's July 2022 guidance states 12:00 CET gate closure and 12:45 CET result publication, consistent with its May 2021 timing change. The issue falls after the normal auction sequence; exceptional

auction delays and historical dissemination receipts were not recovered. The remaining intended role is subsequent operational review and intraday preparation, not participation in that closed auction. Current Energinet price-service guidance also describes retrieval around its daily update; this is current guidance, not reconstructed historical availability.

A slot is a selected delivery-hour window, not one hour of staff labour. Review duration, number of areas checked, bundling of adjacent windows and required follow-up determine actual staff time. Neither staff minutes nor the cost of false operational actions was measured. No operator interviews, expert-panel validation or prospective deployment were conducted. Passive live inference would need to log input vintages and receipt times, issued probabilities, selected windows, review duration and resulting decisions before operational benefit could be tested. Che et al. (2026, Section 4.3) explicitly separates historical evaluation, passive live inference and subsequent operational integration in regional wind-power forecasting.

S13.2. Price data, fixed definitions and completed analysis

This exploratory question was specified after existing forecast and policy results had been inspected, but before downloading or inspecting Danish price values: do the already fixed q90 own-zone and two area events coincide with higher prices for the same delivery hour? The analysis does not change the weather, thresholds, marginals, dependence selection or allocation parameters. Neither prices nor information derived from them enters the forecasting or policy pipelines.

Official Energinet Elspotprices gives hourly EUR/MWh until the end of September 2025 local delivery dates; current metadata directs subsequent records to DayAheadPrices, with 15-minute TimeUTC intervals from 1 October 2025. The metadata defines UTC timestamps as interval starts and identifies DK1 west and DK2 east of the Great Belt. Broad date guard bands were downloaded, then trimmed to 2024-01-01 00 UTC through 2025-12-31 23 UTC. Quarter-hour aggregation requires all four unique 00, 15, 30 and 45 minute records within a UTC hour; their equally weighted arithmetic mean is the hourly delivery price. No missing interval is interpolated. EUR values avoid changes in provider DKK conversion. Each zone has 17,544 unique finite hourly prices across the two calendar years; all 17,400 forecast-score hours in 725 complete UTC days per zone match a finite price. The common score domain has 8,712 hours in 2024 and 8,688 hours in 2025.

The own-zone event is the already fitted seasonal q90 DK exceedance. The joint event is at least two q90 exceedances among DE-LU, France and that Danish zone. A descriptive DK-pivotal subset requires DK exceedance and exactly one DE-LU/French exceedance; it is nested within the joint event and is not an independent validation population. Reported groups are evaluated separately by zone and UTC calendar year. EUR 100/MWh was fixed before price acquisition as a descriptive price cut, not a scarcity, adequacy or reserve threshold. Medians and interquartile ranges are calculated within observed groups. Table S13.1 retains all groups, including non-events and the nested subset.

Table S13.1. Same-delivery-hour Danish day-ahead prices by fixed event and UTC year. Groups use the unchanged primary local-weather score domain. Annual partitions account for the change in overall price levels; they do not condition on other price determinants. Hourly prices after the resolution change are complete four-quarter averages.

Danish same-hour price context by zone, year and fixed event status: supporting-table file **supporting_table_10.csv** (24 rows). The complete Markdown rendering is also retained with the supporting tables. No rows or outcomes were dropped; this separates the audit table from the reading text.

For the own-zone and joint comparisons, exploratory pointwise 95% percentile intervals use 2,000 circular moving-block resamples of 14 UTC days, separately within each year and quarter. Every price and event label is resampled together on the full calendar lattice, preserving missing forecast days as missing rather than filling them. Medians and price-cut fractions are recomputed in each draw. No independent-hour standard

error, p value or familywise significance claim is reported. Intervals describe the variability of these fixed retrospective contrasts, not uncertainty about an operating benefit. The nested DK-pivotal summaries are descriptive only.

Table S13.2. Exploratory event-minus-non-event price differences. Percentages are percentage-point differences; intervals are pointwise 14-day block intervals. These are contemporaneous associations, with no causal adjustment or incremental forecasting claim.

Zone	UTC year	Event definition	Median difference EUR/MWh [95%]	≥100 share difference pp [95%]
DK1	2024	Own-zone q90	34.34 [25.38,41.67]	29.79 [20.02,37.21]
DK1	2024	Joint ≥2 q90	44.08 [37.10,52.16]	42.79 [33.93,51.14]
DK1	2025	Own-zone q90	21.41 [13.00,33.05]	24.88 [15.43,33.89]
DK1	2025	Joint ≥2 q90	45.76 [38.05,55.95]	53.60 [46.29,60.72]
DK2	2024	Own-zone q90	37.76 [29.95,46.17]	32.88 [23.44,40.75]
DK2	2024	Joint ≥2 q90	42.88 [35.55,51.32]	39.66 [31.34,48.67]
DK2	2025	Own-zone q90	32.72 [24.19,40.87]	35.32 [27.42,42.18]
DK2	2025	Joint ≥2 q90	47.66 [41.04,56.64]	54.11 [49.22,59.50]

The comparisons support a market-context link between the monitoring event and higher own-zone prices, rather than a price forecasting result. Price formation can reflect demand patterns, fuel costs, renewable generation, import conditions and congestion. The comparisons do not adjust for those determinants. Current retrieved prices are not a historical first-release archive. Under ordinary market timing, day-ahead auction formation precedes the fixed forecast issue; we did not recover price-release vintages or establish when each value was publicly downloadable. No incremental predictive information relative to already announced prices, warning-induced price response, economic decision value, reserve insufficiency, network constraints or outages is identified.

S13.3. Scale of existing Danish accounting components

The existing settlement records retain gross consumption, commercial self-consumption (LocalPowerSelf-ConMWh) and solar self-consumption. Official metadata warns that the commercial component is uncertain and overlaps the solar self-consumption estimate. The implemented load is gross consumption minus commercial self-consumption; net load then deducts total solar output, including solar self-consumption, and wind. This equals the official through-grid consumption definition, Gross-LocalSelf-SolarSelf, minus grid-delivered PV and wind. SolarSelf is not deducted a second time. A second subtraction would be a bookkeeping error. Table S13.3 describes the current source's component magnitudes for all UTC hours in 2024 and 2025, not only the forecast-score domain. Hourly MWh divided by one hour equals mean MW. Shares are quantiles of the hourly component/gross ratios; a share quantile is not the ratio of separately computed component and gross quantiles. These provider-accounting magnitudes do not estimate a harmonised cross-country bias, and thresholds or events are not refitted for this descriptive check. Historical classification warnings remain, including the 2019 grouping of PV in the self-consumption field.

Table S13.3. Danish current-vintage accounting-component magnitudes. Source is Energinet ProductionConsumptionSettlement. The study does not infer the direction of joint warning error from these accounting differences.

Zone	UTC year	Accounting component	Hours	Median MW	95th percentile MW	Maximum MW	Median hourly share of gross	95th percentile hourly share
DK1	2024	Commercial self-consumption	8784	47.85	123.05	169.85	1.85%	4.16%
DK1	2024	Solar self-consumption	8784	1.24	125.10	445.55	0.05%	4.07%
DK1	2025	Commercial self-consumption	8760	68.69	206.76	294.07	2.40%	6.32%
DK1	2025	Solar self-consumption	8760	1.20	100.69	331.75	0.04%	3.07%
DK2	2024	Commercial self-consumption	8784	16.37	47.89	67.87	0.90%	2.81%
DK2	2024	Solar self-consumption	8784	0.30	90.74	226.72	0.02%	5.19%
DK2	2025	Commercial self-consumption	8760	17.57	62.85	88.89	0.96%	3.40%
DK2	2025	Solar self-consumption	8760	0.36	102.18	298.62	0.02%	5.41%

S13.4. Source locators and reproducibility boundary

Price-source definitions and the transition are verified against official metadata: <https://api.energidataservice.dk/meta/dataset/Elspotprices> and <https://api.energidataservice.dk/meta/dataset/DayAheadPrices>. Gross and self-consumption definitions are documented at <https://api.energidataservice.dk/meta/dataset/ProductionConsumptionSettlement>. Redistribution remains subject to the source terms at <https://www.energidataservice.dk/terms-and-conditions>. The operating context uses ENTSO-E's OPC/STA and Regions pages and Nord Pool's 2022 auction guidance and 2021 timing notice. These links and dated bibliographic references are provided with the manuscript references.

The analysis retains a plan written before price acquisition, acquisition receipts, unchanged forecast-source checksums, normalised hourly prices, fixed event joins, group summaries, block intervals and an independent summary readback. All 24 summary groups and eight point differences were independently recomputed from the joined hourly table; native quarter-hour completeness and raw-file checksums were verified. These records support computational reproducibility of current-vintage association, not first-release reconstruction. No expert feedback or live use is claimed, and the responsibilities of the human authors for the final manuscript, the sources and the analysis remain as stated in the AI assistance statement of the main article.

S14. Reproduction and release scope

The accompanying reproducibility package contains a tested aggregate-only analysis subset, the source pipeline, fixed configuration files, main result tables, six new scientific figures and complete supporting tables. Raw source archives, account credentials, installed runtimes and downloaded article texts are excluded. The analysis subset reconstructs aggregate scores, matched coverage and finite-family inference from retained numerical inputs; this should not be confused with independently retraining every forecast from raw archives. The full forecasting source and data-acquisition requirements are separately identified.

The package's reproduction map links each main table and figure to its script and inputs. Run commands and environment requirements are in its README. Figure captions identify pointwise intervals, native outcomes, workload-standardised expectations and post hoc cases. Figure S1 retains the original illustrative priority cases; it is not an additional validation sample.

The package is openly deposited in Zenodo as Cross-border joint high net load warnings: research code and derived evaluation data, version 1.0.0, DOI 10.5281/zenodo.23100589, which is the record reported in the Data Availability Statement of the main article. The licence and the terms of reuse are those stated in that record. Availability of an open access article remains distinct from permission to redistribute every provider archive, and the excluded items are listed above.

S15. Supplementary figure

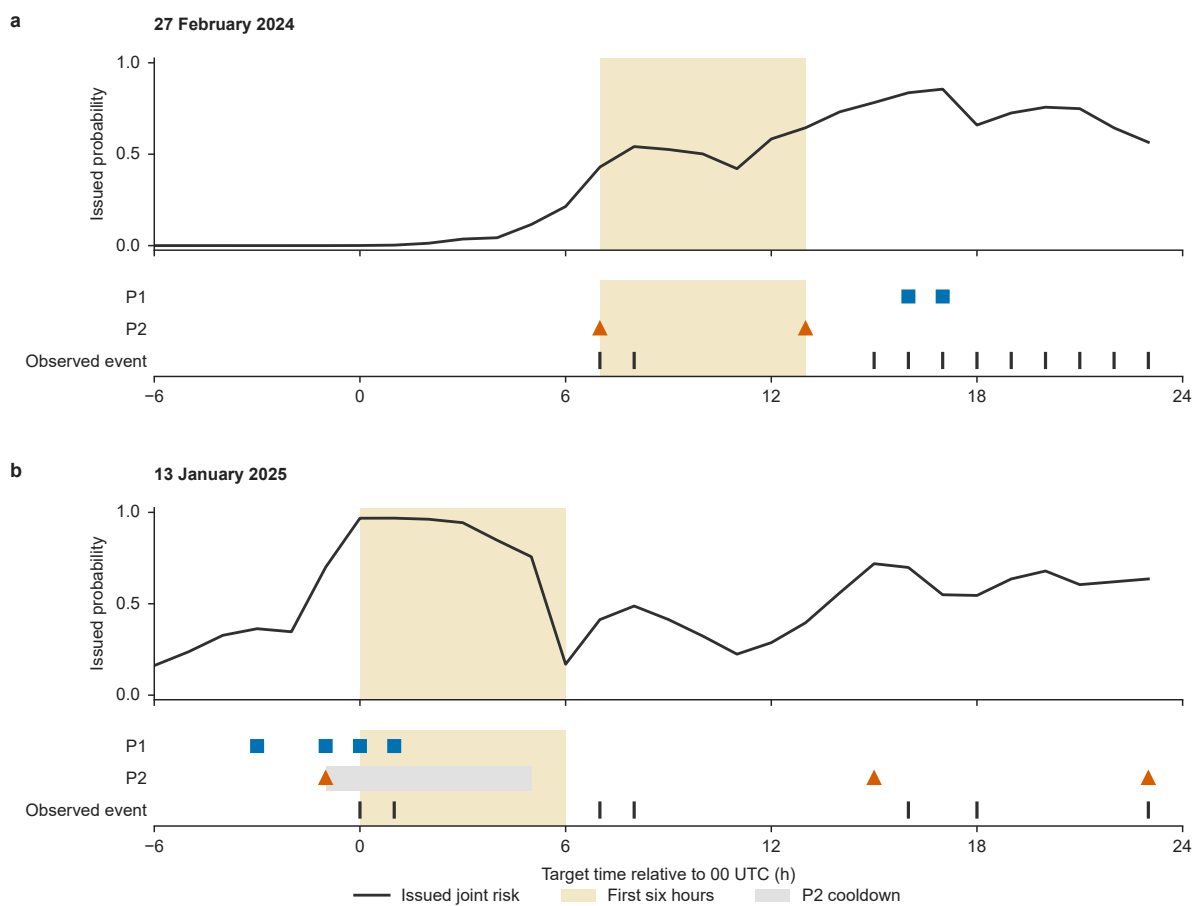


Fig. S1. Post hoc examples of gains and losses from priority on rising risk. The same two previously selected cases illustrate directional differences between P1 and P2. Upper panels show issued joint risk, and lower lanes separate forecast-only planned hours and realised event hours. The tan bands mark the first six elapsed hours after observed onset. The grey P2 band in the second case marks the cooldown after a planned hour at minus one hour. Time is relative to 00 UTC on the date shown. Observed onset and events annotate the cases and do not trigger allocation. These illustrations are not independent predictive validation.